

### 3.1 Introduction

Friction is the phenomenon by which a tangential shearing force is required in order to displace two contacting surfaces along a direction parallel to their nominal contact plane. The study of friction is of great importance in rock mechanics. Its effects arise on all scales: (i) on the microscopic scale, in which friction occurs between faces of minute “Griffith” cracks, (ii) on a somewhat larger scale in which it occurs between individual grains or pieces of aggregate, and (iii) on an even larger scale, on the order of many square meters, over which friction between fault surfaces occurs.

Modern knowledge of friction began with the work of Amontons in France at the end of the seventeenth century. Much experimental work on friction between solid surfaces, and theoretical attempts to explain friction in terms of the geometry and physical properties of the contacting surfaces, is summarized in the monograph of Bowden and Tabor (1985). More recently, technical advances such as atomic force microscopy have yielded new insights into the molecular basis of solid–solid friction (Krim, 1996). However, most of this understanding has been developed for metals and other engineering solids, and its applicability to rock surfaces remains uncertain. Hence, although an atomic-level explanation of rock friction remains elusive, many aspects of the frictional behavior of rock surfaces are indeed understood on a phenomenological level.

In §3.2, the classical theory of friction between solid (typically metallic) surfaces is discussed. Friction on rock surfaces is treated in §3.3. The interesting phenomenon of “stick–slip” between two sheared surfaces in contact is discussed in §3.4. The sliding of two adjacent pieces of rock along a contact surface or indeed any “plane of weakness” is analyzed in some detail in §3.5. Finally, §3.6 treats some aspects of rock friction, namely the effects of time and velocity, which are not accounted for in the simple classical theory.

Frictional forces act along faces of cracks, fractures, and faults in rock. These structures are discussed in more detail in later chapters. However, the basic ideas of rock friction present a simple context in which to utilize the Mohr’s circle construction. They also serve as a starting point for the development of failure laws for intact rock. Finally, in terms of historical development, friction was perhaps the first aspect of rock behavior to be studied and partially understood.

For these reasons, it seems fitting to place the treatment of friction at this early point in the text.

### 3.2 Amontons' law

Suppose that two bodies in contact over an approximately planar surface are pressed together by a normal force  $N$  (Fig. 3.1a). The apparent (macroscopic, nominal) area of contact is  $A$ . If a tangential shearing force is applied parallel to the plane of contact, it is observed that this force must reach some critical value  $T$  in order for sliding to occur. The relationship between  $T$  and  $N$  may be written as

$$T = \mu N, \quad (3.1)$$

where  $\mu$  is called the *coefficient of friction*. This coefficient will depend on the nature of the two surfaces. It might also be expected to depend on the normal load  $N$ , and/or on the nominal surface area  $A$ . However, to a good approximation, in many cases this coefficient is independent of both the normal load and the apparent area of contact – an empirical observation that is known as *Amontons' law*, first enunciated in 1699. In this case the shearing force needed to cause relative motion between the two surfaces is directly proportional to the normal force pressing the surfaces together. Dividing both sides of (3.1) by the nominal area  $A$  yields

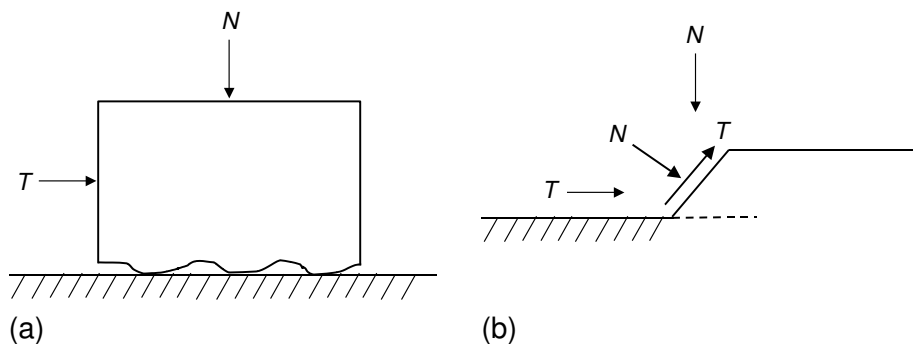
$$\tau = \mu \sigma, \quad (3.2)$$

where  $\tau$  is the shear component of the traction acting along the contact plane, and  $\sigma$  is the normal component. These are usually somewhat loosely referred to as the *shear stress* and *normal stress* acting on the contact plane.

Amontons had in fact found that  $\mu$  was also independent of the materials and the surface finish of the contact region, with a value of about  $1/3$ . More accurate experiments by other researchers later showed that  $\mu$  may vary from values close to 0, for Teflon–Teflon contacts, to values as high as 1.5, for nickel–nickel contacts (Bowden and Tabor, 1985).

Equation (3.2) refers to the condition necessary to initiate sliding, starting from a state of rest. Hence,  $\mu$  is more properly referred to as the coefficient of *static* friction. Once sliding has been initiated, however, it is found that the

**Fig. 3.1** Two bodies in contact along (a) a nominally planar surface, and (b) a surface having a local region inclined at an angle  $\theta$ , acted on by a normal force  $N$  and a tangential force  $T$ .



shear stress required to maintain motion is somewhat less than  $\mu\sigma$ . In fact, the condition for maintaining relative motion at constant velocity is  $\tau = \mu_d\sigma$ , where  $\mu_d < \mu$  is the coefficient of *dynamic* friction. For many engineering materials,  $\mu_d$  is found to be nearly independent of the relative velocity of the two surfaces – an approximate “law” that was first noted by the French physicist Coulomb in 1785.

For irregular surfaces, it may happen that sliding takes place up a plane inclined by some small angle  $\theta$  to the nominal plane of contact (Fig. 3.1b). The normal and tangential forces acting on this inclined surface are

$$N' = N \cos \theta + T \sin \theta, \quad T' = T \cos \theta - N \sin \theta. \quad (3.3)$$

If sliding along the incline is governed by  $T' = \mu N'$ , then the two force components acting along the nominal plane of contact are related by

$$T = N(\tan \theta + \mu)/(1 - \mu \tan \theta). \quad (3.4)$$

For small angles,

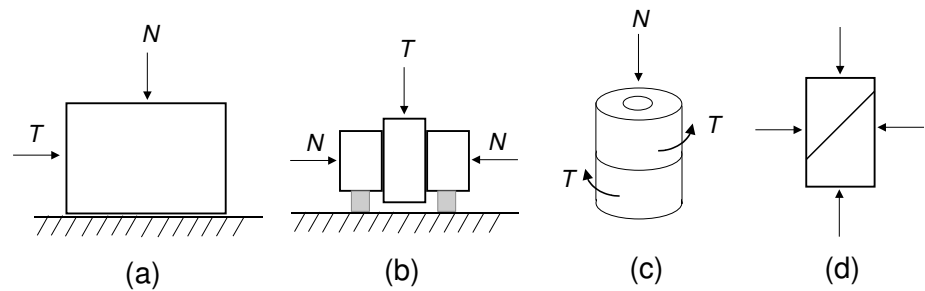
$$T = [\mu + (1 + \mu^2)\theta]N, \quad (3.5)$$

so that there is an increase in the *apparent* coefficient of friction, by an amount proportional to  $\theta$ .

### 3.3 Friction on rock surfaces

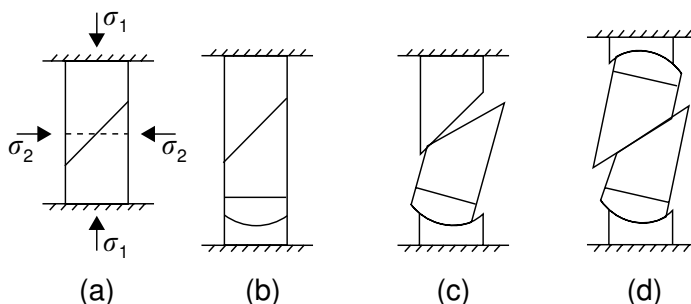
Frictional effects are of importance in rock mechanics mainly in two connections. On a small scale, friction may occur between the facing surfaces of microcracks. On a larger scale, friction occurs between the surfaces of joints or fracture planes. The surfaces in question may be new surfaces, such as along a tension joint on which no sliding has yet occurred, or they may be old surfaces on which considerable relative motion has already taken place. The scale and condition of the surfaces on which laboratory measurements are made will usually lie somewhere between these two extremes.

Measurements of rock friction can be made using any of the configurations shown in Fig. 3.2. In configuration (a), essentially the same as shown in Fig. 3.1a, the two rock surfaces are pressed together by a normal force  $N$  and sheared by a tangential force  $T$  (Penman, 1953; Bowden, 1954; Horn and Deere, 1962). In practice, this configuration requires elaborate precautions to ensure that the normal load is applied uniformly across the surface. Hoskins et al. (1968) used



**Fig. 3.2** Several configurations used in measuring friction along rock surfaces.

**Fig. 3.3** Different configurations used for measuring friction in a triaxial testing machine.



configuration (b), with the normal forces  $N$  applied by flat pressure cells and the tangential force  $T$  applied by a standard rock-testing machine (§6.4). In this configuration, the amount of contact area will be constant, but the contact will not occur over precisely the same surfaces throughout the process. The rotary system (c) has the advantage that the same surfaces will be in contact, regardless of the amount of displacement that has occurred.

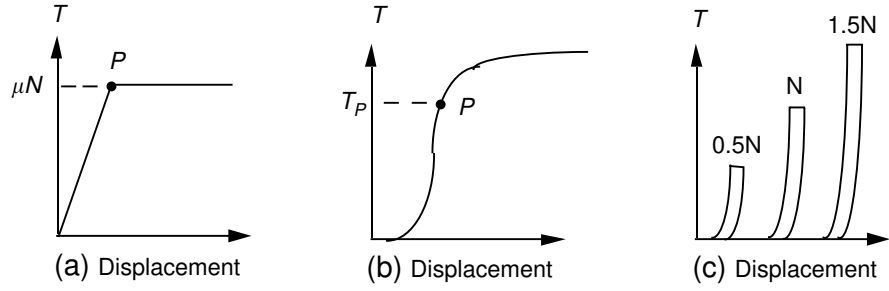
The most commonly used configuration is (d), in which a cylindrical core is sliced at an angle  $\beta$  to the diametral plane, and placed in a standard “triaxial” testing cell. The specimen is jacketed in rubber or copper, lateral pressure  $\sigma_2 = \sigma_3$  is applied to its lateral curved surface by oil pressure, and an axial stress  $\sigma_1$  is applied by the testing machine. This method allows high normal and tangential stresses to be applied to the sample, but has the disadvantage that a change in geometry, or stress, or both, occurs after any slippage between the two surfaces.

Three variations of this configuration may be used (Fig. 3.3). In configuration (a), the cylindrical specimen is situated between rigid platens whose faces are perpendicular to the maximum principal stress  $\sigma_1$ . After initial slip has occurred, a lateral stress is produced whose magnitude is not known. In (b), a single spherical seat is used, in which case the specimen rotates into configuration (c) as slip occurs, after which contact no longer exists over the original planar surfaces of elliptical cross section. If two spherical seats are used, as in (d), the specimen halves will rotate so as to maintain the angle of contact. However, the amount of contact area will decrease as slip proceeds, and lateral forces are again produced. Hence, in each configuration, either the geometry or the forces change as slippage occurs. Nevertheless, for small amounts of slip, consistent results can in fact be obtained using all three methods.

Jaeger (1959), Byerlee (1967) and others have used the method of a single spherical seat, (b). The experiments can be conducted using any angle  $\beta$  between about  $30^\circ$  and  $65^\circ$ . The inferred values of  $\mu$  might be expected to vary with  $\beta$ , but in fact are found to be nearly independent of the angle, indicating that the changes in geometry shown in Fig. 3.3c do not greatly contaminate the results.

Consider again the configuration shown in Fig. 3.2b. According to the simple classical theory as expressed by Amontons’ law, no sliding takes place when  $T < \mu N$ , although a small elastic deformation of the central block will occur. After  $T$  reaches the value  $\mu N$ , slippage will commence, starting at point  $P$ . If this slippage is constrained to occur at constant velocity, the tangential force will be

**Fig. 3.4** (a) Force–displacement curve for idealized frictional behavior. (b) Schematized force–displacement curve obtained by Hoskins et al. (1968) for a moderately rough trachyte surface in an apparatus similar to that of Fig. 3.2b. (c) Subsequent force–displacement curves for the same surfaces, under normal loads that are 0.5, 1.0, and 1.5 times that used in (b).



$T = \mu_d N$ , where  $\mu_d$  is the *coefficient of dynamic friction*. This idealized behavior is shown schematically in Fig. 3.4a, where, for simplicity, we take  $\mu_d = \mu$ .

An example of actual data obtained by Hoskins et al. (1968) on a moderately rough trachyte surface is shown in Fig. 3.4b, although the results are typical of those found for other rocks. The initial elastic displacement is seen to be slightly nonlinear. Sliding begins at  $P$ , and if the coefficient of friction is defined based on the point of first slippage, we would find  $\mu = T_P/N$ . But the tangential force needed to produce continued sliding in fact increases, approaching an asymptotic value after a displacement of about 2 cm. This increase is attributable to the development of a new surface of sliding, which contains slickensides and detrital material. If sliding is reinitiated under a larger value of  $N$ , a constant value of  $T$  is attained more rapidly (Fig. 3.4c).

If the data obtained by Hoskins et al. (1968) are interpreted in accordance with (3.3), in which  $\mu = T/N = \tau/\sigma$ , it is found that the coefficient of friction decreases with increasing normal stress, stabilizing at high values of  $\sigma$  (Fig. 3.5b). However, if  $\tau$  is plotted against  $\sigma$ , as in Fig. 3.5a, the data fall on nearly straight lines. This suggests using the following equation to describe rock friction:

$$\tau = S_o + \mu\sigma, \quad (3.6)$$

which was originally proposed by Coulomb in 1785. Borrowing the terminology used in soil mechanics,  $S_o$  is called the *cohesion* of the surface, and  $\mu$  the *coefficient of friction*. As this equation gives a different definition of  $\mu$  than does (3.3), we henceforth use  $\mu^*$  to refer to the ratio  $\tau/\sigma$ . Thus,  $\mu^*$  is the secant of the  $\tau$ – $\sigma$  curve, and  $\mu$  is the (best-fit) tangent of that curve. If (3.6) is indeed obeyed, these two friction coefficients are related by  $\mu^* = \mu + (S_o/\sigma)$ .

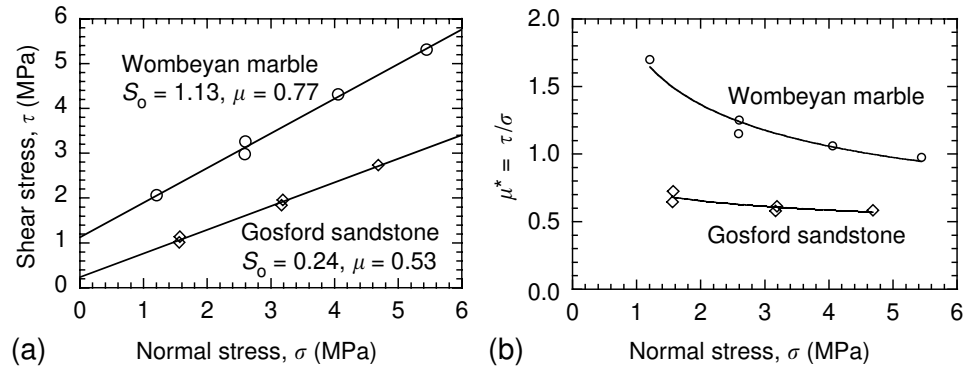
Values of the friction coefficient  $\mu$  of various rocks and minerals are shown in Table 3.1. The measured values vary with rock type, and also depend on surface finish and whether or not the surface is wet or dry. The presence of water increases  $\mu$  in some cases, decreases it in others, and in some cases has little effect.

Byerlee (1978) found that the frictional behavior of a broad range of common rock types could be fit by the following bilinear empirical expression:

$$\begin{aligned} \tau &= 0.85\sigma \quad \text{for } \sigma < 200 \text{ MPa}, \\ \tau &= 50 \text{ MPa} + 0.6\sigma \quad \text{for } 200 < \sigma < 1700 \text{ MPa}, \end{aligned} \quad (3.7)$$

which is often referred to as *Byerlee's law*. This law is equivalent to taking a different pair of values for  $S_o$  and  $\mu$  in the two ranges  $\{\sigma < 200 \text{ MPa}\}$  and

**Fig. 3.5** (a) Shear stress plotted against normal stress for two rock surfaces. (b) Same data, with  $\mu^* = \tau/\sigma$  plotted against  $\sigma$  (Hoskins et al., 1968).



**Table 3.1**

Coefficient of friction of some rocks and minerals.

*Notation:*

B = Bowden and Tabor (1985),  
By = Byerlee (1967),  
H = Hoskins et al., (1968), HS = Handin and Stearns (1964),  
HD = Horn and Deere (1962),  
J = Jaeger (1959),  
P = Penman (1953),  
R = Rae (1963);  
l = large surface,  
s = small surface,  
t = triaxial test,  
r = rough surface,  
c = coarsely ground surface, f = finely ground surface,  
n = natural shear surface, w = wet surface, d = dry surface.

| Mineral                              | $\mu$ | Mineral         | $\mu$ | $\mu$ (wet) |
|--------------------------------------|-------|-----------------|-------|-------------|
| NaCl [B,s]                           | 0.7   | Quartz [HD]     | 0.11  | 0.42        |
| PbS [B,s]                            | 0.6   | Quartz [P]      | 0.19  | 0.65        |
| S [B,s]                              | 0.5   | Feldspar [HD]   | 0.11  | 0.46        |
| Al <sub>2</sub> O <sub>3</sub> [B,s] | 0.4   | Calcite [HD]    | 0.14  | 0.68        |
| Ice [B,s]                            | 0.5   | Muscovite [HD]  | 0.43  | 0.23        |
| Glass [B,s]                          | 0.7   | Biotite [HD]    | 0.31  | 0.13        |
| Diamond [B,s]                        | 0.1   | Serpentine [HD] | 0.62  | 0.29        |
| Diamond [B,s,c]                      | 0.3   | Talc [HD]       | 0.36  | 0.16        |

| Rock                 | $\mu$ | Mineral            | $\mu$ |
|----------------------|-------|--------------------|-------|
| Sandstone [R]        | 0.68  | Trachyte [H,l,f]   | 0.63  |
| Sandstone [J,t,n]    | 0.52  | Trachyte [H,l,c]   | 0.68  |
| Sandstone [H,l,r]    | 0.51  | Trachyte [H,l,c,w] | 0.56  |
| Sandstone [H,l,r,w]  | 0.61  | Marble [H,l,f]     | 0.75  |
| Granite [By,t,n,c]   | 0.60  | Marble [J,t,n]     | 0.62  |
| Granite [By,t,n,c,w] | 0.60  | Porphyry [J,t,n]   | 0.86  |
| Granite [H,l,g]      | 0.64  | Dolomite [H,s,t,c] | 0.40  |
| Gneiss [J,t,n]       | 0.71  | Gabbro [H,l,f]     | 0.18  |
| Gneiss [J,t,n,w]     | 0.61  | Gabbro [H,l,c]     | 0.66  |

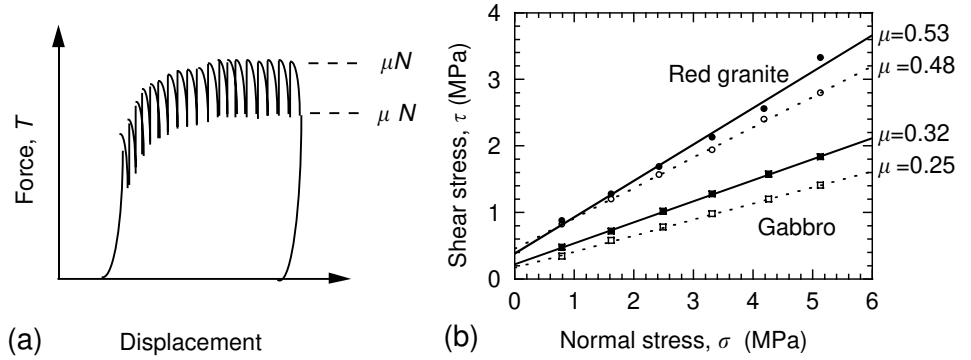
$\{\sigma > 200 \text{ MPa}\}$ . Notable exceptions to this law include clays and other sheet silicates (Lockner, 1995).

### 3.4 Stick-slip oscillations

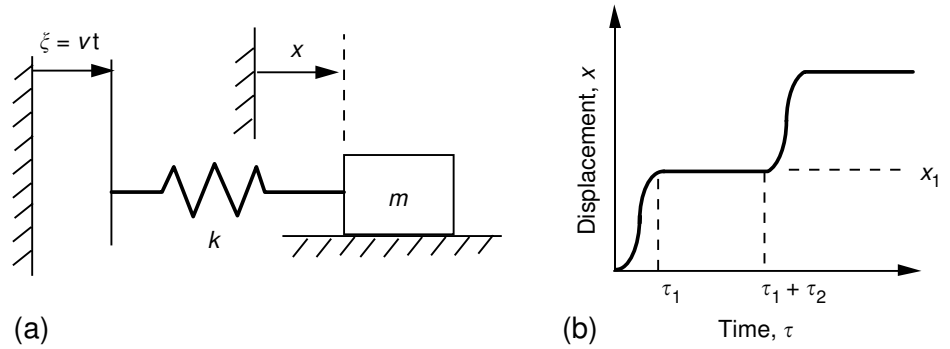
If rock friction tests are conducted on finely ground or polished surfaces, a phenomenon known as “stick-slip” is sometimes found to occur (Fig. 3.6a). If a shear load  $T$  is applied as the normal load  $N$  is held constant, the displacement increases gradually as the shear force  $T$  increases, until at some point it jumps abruptly, after which the surfaces lock together again, and the force drops to a lower value. This process continues, with each sticking phase followed by a slip

**Fig. 3.6**

(a) Force–displacement for a granite surface in the apparatus of Fig. 3.2b, under different normal loads. (b) Variation of maximum and minimum shear stress as a function of the normal load (Hoskins et al., 1968).



**Fig. 3.7** (a) Model of a simple mechanical system that will exhibit stick-slip oscillations. (b) Motion of this system as a function of time.



phase of very short duration. After several cycles the maximum and minimum values of the shear load  $T$  each reach asymptotic values. If these asymptotic values of  $T_{\max}$  and  $T_{\min}$  are plotted against  $N$ , straight lines of the form

$$|\tau| = S_0 + \mu\sigma, \quad \tau = S'_0 + \mu'\sigma, \quad (3.8)$$

are obtained (Fig. 3.6b). As  $\mu$  is obtained from the values of  $T$  that occur immediately before slip occurs, it represents the coefficient of *static* friction.

A simple model for a system exhibiting stick-slip oscillations can be formulated that requires only the assumption that  $\mu_d < \mu$ . Consider a body of mass  $m$  resting on a horizontal surface, attached on one side to a spring having stiffness  $k$  (Fig. 3.7a). The normal force exerted by the mass on the surface is  $mg$ . (An additional normal load  $N$  could be assumed to be pressing the mass against the surface, but this adds an additional parameter into the equations without giving qualitatively different behavior.) Let the free end of the spring be moved to the right at a constant velocity  $v$ , in which case the position of the free end is given by  $\xi = vt$ . If the position  $x$  of the mass is measured from its initial location, the spring is compressed by an amount  $\xi - x = vt - x$ , and the rightward force exerted by the spring on the mass will be  $T = k(vt - x)$ . If the mass is initially at rest, it will not move until the force  $T$  reaches  $\mu mg$ , which will occur at time  $t^* = \mu mg/kv$ . It is convenient to use the elapsed time as the independent variable since the initial movement of the mass, that is,  $\tau = t - t^*$ , so that  $T = k(v\tau + vt^* - x)$ .

At subsequent times, the mass will be in motion, and this motion will be resisted by a frictional force,  $F = -\mu_d mg$ . A horizontal force balance on the

mass gives

$$m\ddot{x} = k(v\tau + vt^* - x) - \mu_d mg, \quad (3.9)$$

where the overdot denotes differentiation with respect to the elapsed time,  $\tau$ . The solution to (3.9), for the initial conditions  $x = \dot{x} = 0$  when  $\tau = 0$ , is

$$x = v\tau - (v/\omega) \sin \omega\tau - [(\mu - \mu_d)g/\omega^2] \cos \omega\tau + (\mu - \mu_d)g/\omega^2, \quad (3.10)$$

$$\dot{x} = v - v \cos \omega\tau + [(\mu - \mu_d)g/\omega] \sin \omega\tau. \quad (3.11)$$

where  $\omega = \sqrt{k/m}$  is the undamped natural frequency of the system, and where use has been made of the fact that  $kvt^* = \mu mg$ .

The mass will move to the right in accordance with (3.10), coming to rest when  $\dot{x} = 0$ . From (3.11), the “slip” phase will therefore end when

$$-(\mu - \mu_d)g/\omega v = (1 - \cos \omega\tau_1)/\sin \omega\tau_1 \equiv \tan(\omega\tau_1/2), \quad (3.12)$$

which can be inverted to give the duration of the slip phase,

$$\tau_1 = \frac{2\pi}{\omega} - \frac{2}{\omega} \tan^{-1} \left[ \frac{(\mu - \mu_d)g}{\omega v} \right]. \quad (3.13)$$

The displacement at this time is found from (3.10), (3.12), and (3.13) to be given by

$$x_1 \equiv x(\tau_1) = v\tau_1 + 2(\mu - \mu_d)g/\omega^2, \quad (3.14)$$

and the force exerted on the mass by the spring at this time will be

$$T_1 \equiv T(\tau_1) = k(v\tau_1 + vt^* - x_1) = \mu mg - 2k(\mu - \mu_d)g/\omega^2 = (2\mu_d - \mu)mg. \quad (3.15)$$

Noting that  $\mu_d < \mu$ , the spring force at time  $\tau_1$  will be less than  $\mu mg$ , and so the mass will be at rest. After this time, the spring will continue to compress, but the mass will be in its “sticking” phase. Hence,  $T_1$  represents the *minimum* spring force that occurs at any time during the process, and so comparison with (3.8) shows that  $2\mu_d - \mu = \mu'$ , implying that the dynamic friction coefficient is equal to the mean value of  $\mu_{\max} = \mu$  and  $\mu_{\min} = \mu'$ .

Slip will recommence when the spring force again reaches  $\mu mg$ . This will occur after an additional *elapsed* time of

$$\tau_2 = 2(\mu - \mu_d)mg/kv = 2(\mu - \mu_d)g/\omega v, \quad (3.16)$$

after which the cycle will repeat (Fig. 3.7b). The duration of the stick phase is  $\tau_2$ , and so the stick–slip cycle will have a total period of  $\Delta\tau = \tau_1 + \tau_2$ . Comparison of (3.13) and (3.16) shows that

$$(\omega\tau_1/2) = \pi - \tan^{-1}(\omega\tau_2/2). \quad (3.17)$$

The usual situation is that the velocity  $v$  is “small,” in the sense that  $v \ll (\mu - \mu_d)g/\omega$ , in which case (3.13) and (3.16), or (3.17), imply that  $\tau_1 \ll \tau_2$ . Hence,



from (3.14), the displacement occurring over each cycle will be approximately given by

$$\Delta x \approx 2(\mu - \mu_d)g/\omega^2 = 2(\mu - \mu_d)mg/k. \quad (3.18)$$

But the maximum and minimum spring forces during each cycle are  $T_{\max} = \mu mg$  and  $T_{\min} = (2\mu_d - \mu)mg$ , so

$$\Delta x(\text{cycle}) = (T_{\max} - T_{\min})/k. \quad (3.19)$$

Brace and Byerlee (1966) suggested that stick-slip between opposing faces of a fault may provide a mechanism for earthquakes. In this context, (3.19) would give a relationship between the amount of slip along a fault, the stress drop associated with that slip, and the “stiffness” of the surrounding rock mass. Burridge and Knopoff (1967) generalized the model by considering a chain of masses  $\{m_1, m_2, \dots, m_N\}$ , connected in series by a set of springs with stiffnesses  $\{k_1, k_2, \dots, k_N\}$ . This model predicts chaotic sequences consisting of many small slips and occasional large ones, which bear a resemblance to earthquake sequences. Further developments along these lines have been made by Mora and Place (1994) and others.

### 3.5 Sliding on a plane of weakness

In two dimensions, suppose that the rock has a preexisting plane of weakness whose outward unit normal vector makes an angle  $\beta$  with the direction of the maximum principal stress,  $\sigma_1$  (Fig. 3.8a). The criterion for slippage to occur along this plane is assumed to be

$$|\tau| = S_o + \mu\sigma, \quad (3.20)$$

as in (3.6), where  $\sigma$  is the normal traction component acting along this plane, and  $\tau$  is the shear component. By (2.54) and (2.55),  $\sigma$  and  $\tau$  are given by

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_2) + \frac{1}{2}(\sigma_1 - \sigma_2) \cos 2\beta, \quad (3.21)$$

$$\tau = -\frac{1}{2}(\sigma_1 - \sigma_2) \sin 2\beta. \quad (3.22)$$

These expressions may also be written as

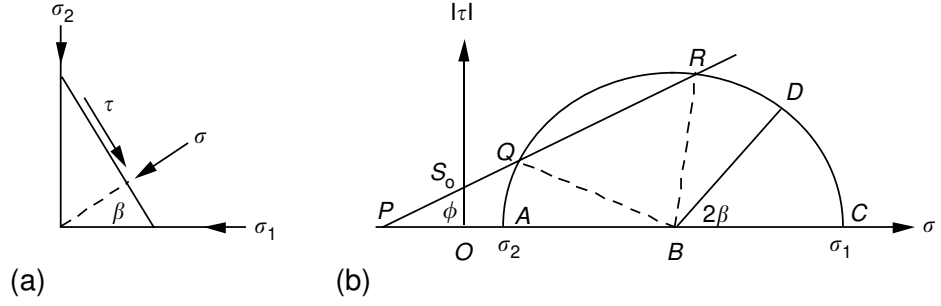
$$\sigma = \sigma_m + \tau_m \cos 2\beta, \quad \tau = -\tau_m \sin 2\beta, \quad (3.23)$$

where  $\sigma_m = (\sigma_1 + \sigma_2)/2$  is the mean normal stress, and  $\tau_m = (\sigma_1 - \sigma_2)/2$  is the maximum shear stress.

If we define the *angle of internal friction*,  $\phi$ , by the relation

$$\mu = \tan \phi, \quad (3.24)$$

**Fig. 3.8** (a) Plane of weakness with outward normal vector oriented at angle  $\beta$  to the direction of maximum principal stress. (b) Situation described on a Mohr diagram.



then condition (3.20) can be written in the following equivalent forms:

$$\tau_m(\sin 2\beta - \tan \phi \cos 2\beta) = S_0 + \sigma_m \tan \phi, \quad (3.25)$$

$$\tau_m = (\sigma_m + S_0 \cot \phi) \sin \phi \operatorname{cosec}(2\beta - \phi). \quad (3.26)$$

$$\sigma_1[\sin(2\beta - \phi) - \sin \phi] - \sigma_2[\sin(2\beta - \phi) + \sin \phi] = 2S_0 \cos \phi, \quad (3.27)$$

$$\sigma_1 - \sigma_2 = \frac{2(S_0 + \mu \sigma_2)}{(1 - \mu \cot \beta) \sin 2\beta}, \quad (3.28)$$

$$\sigma_1 = \frac{2S_0 \cos \phi}{(1 - k) \sin(2\beta - \phi) - (1 + k) \sin \phi}, \quad (3.29)$$

where  $k = \sigma_2/\sigma_1 \leq 1$ . The stress difference that would be required to cause slippage can be found from (3.28), as a function of  $\beta$ , for a fixed value of the minor principal stress  $\sigma_2$  (Fig. 3.9). If the major principal stress is aligned with the plane of weakness, then  $\beta \rightarrow \pi/2$ , and (3.28) shows that  $\sigma_1 - \sigma_2 \rightarrow \infty$ . The stress difference required to cause slippage also becomes infinite as  $\beta \rightarrow \phi$ . As the right-hand side of (3.28) must by definition be nonnegative, solutions can exist only for  $\phi < \beta < \pi/2$ . By differentiating (3.31), the minimum value of  $\sigma_1$  needed to cause slippage is found to occur when

$$\tan 2\beta = -1/\mu, \quad (3.30)$$

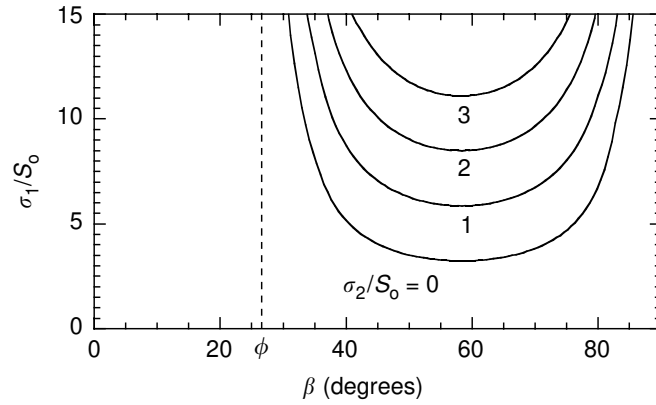
and this minimum value is found to be

$$\sigma_1 = \sigma_2 + 2(S_0 + \mu \sigma_2)[(1 + \mu^2)^{1/2} + \mu]. \quad (3.31)$$

This problem can be analyzed with the aid of the Mohr diagram, Fig. 3.8b. The condition (3.20) for failure is represented by the straight line  $PQR$  that is oriented at angle  $\beta$  to the  $\sigma$ -axis, and intersects the  $|\tau|$ -axis at  $S_0$  and the  $\sigma$ -axis at  $-S_0 \cot \phi$ . The normal and shear tractions along the plane of weakness are represented by the point  $D$  in the  $\sigma$ - $\tau$  plane. Slippage will not occur if the point  $D$  lies within either of the arcs  $AQ$  or  $RC$  of the Mohr circle, but will occur if  $D$  lies within arc  $QR$ . The limiting condition for slippage to occur can be found by imagining  $D$  to be located at either  $Q$  or  $R$ . Taking  $D$  to coincide with  $R$ , the angle  $RBC$  will equal  $2\beta$ , and so the angle  $PRB$  will equal  $2\beta - \phi$ . Applying the law of sines to triangle  $PRB$  shows

$$\frac{BR}{\sin \phi} = \frac{PB}{\sin(2\beta - \phi)}. \quad (3.32)$$

**Fig. 3.9** Variation of  $\sigma_1$  needed to cause sliding on a plane of weakness, for  $\mu = 0.5$ .



But  $|BR| = \tau_m$ , and  $|PB| = |PO| + |OB| = S_o \cot \phi + \sigma_m$ , so

$$\tau_m \sin(2\beta - \phi) = (\sigma_m + S_o \cot \phi) \sin \phi, \quad (3.33)$$

which is equivalent to (3.26). The same result would be obtained by imagining  $D$  to coincide with  $Q$ .

For a given state of stress and a given value of  $\mu$ , there are two values of  $\beta$  that will satisfy (3.33), corresponding to whether  $D$  coincides with  $R$  or  $Q$ . These values are

$$2\beta_1 = \phi + \sin^{-1}\{[(\sigma_m + S_o \cot \phi)/\tau_m] \sin \phi\}, \quad (3.34)$$

$$2\beta_2 = \pi + \phi - \sin^{-1}\{[(\sigma_m + S_o \cot \phi)/\tau_m] \sin \phi\}. \quad (3.35)$$

Sliding will occur on any plane for which  $\beta_1 \leq \beta \leq \beta_2$ .

If the cohesion  $S_o$  were set to zero, the slippage line (3.20) would move to the right by the amount  $|PO| = S_o \cot \phi = S_o/\mu$ . If, at the same time,  $S_o/\mu$  were added to each of the two principal stresses, the Mohr circle would translate to the right by this same amount, and the relative orientation of the Mohr circle and the slippage line would remain unchanged. Hence, results for the case of principal stresses  $\sigma_1$  and  $\sigma_2$ , with a finite value of  $S_o$ , are identical to those for the case of principal stresses  $\sigma_1 + (S_o/\mu)$  and  $\sigma_2 + (S_o/\mu)$ , with  $S_o = 0$ .

If the joint is filled with pore fluid under a pressure  $p$ , it is experimentally found that the theory will continue to apply if the amount  $p$  is first subtracted from each of the principal stresses (Byerlee, 1967). The values  $\sigma_1 - p$ , etc., are often referred to in this context as the *effective principal stresses*.

The basic theory outlined above has three fundamental applications of great importance. The most obvious is the study of sliding across an open joint or cut rock surface for which the criterion for slip has been experimentally found to be given by (3.6). If a joint is filled with a weaker material than that comprising the adjacent rock, the same theory will apply, with  $S_o$  being the cohesion and  $\mu$  the coefficient of internal friction of the infilling material, in the sense of the Coulomb theory of failure described in §4.5. Finally, this theory will also describe the behavior of an anisotropic material possessing parallel planes of weakness.

### 3.6 Effects of time and velocity

Although most rocks obey a Coulomb-type friction law to a first approximation, careful experiments have revealed that the apparent friction coefficient varies with parameters such as time and velocity. Dieterich (1972) performed experiments in which  $\tau$  and  $\sigma$  were held constant for an interval of time  $t$ , after which the shear stress was rapidly increased to the level required to produce slip. In this manner, the coefficient of static friction was measured as a function of the total “stick” time. Results for granite, greywacke, quartzite, and sandstone, under different levels of normal stress, all showed a small but measurable increase in  $\mu$  as a function of the time of stick (Fig. 3.10a). The data from each experiment could be fit with an equation of the form

$$\tau = \mu\sigma \quad \mu = \mu_0 + a \ln t. \quad (3.36)$$

The parameter  $\mu_0$  was found to be in the range of 0.7–0.8, and decreased slightly as the normal stress was increased. The dimensionless parameter  $a$ , which is the rate of increase in  $\mu$  with the natural logarithm of the stick time, was found to be insensitive to normal stress, and took on values of 0.0096, 0.0052, 0.0087, and 0.0069 for granite, greywacke, quartzite, and sandstone, respectively.

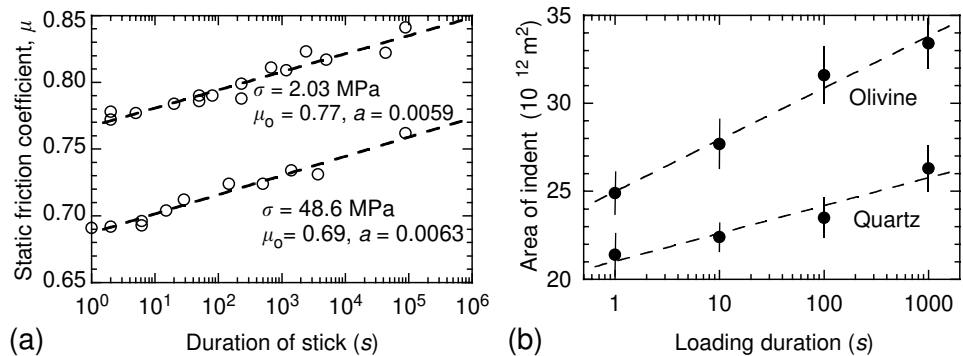
Equation (3.36) fits the data well, for stick times ranging from 1 to  $10^5$  s. Although stick times smaller than 1 s are probably not of physical interest, (3.36) nevertheless predicts unrealistic *negative* values of  $\mu$  as  $t$  approaches zero. To remedy this, Dieterich (1978) proposed to represent the variation of  $\mu$  with  $t$  by

$$\mu = \mu_0 + a \ln[1 + (t/t_0)], \quad (3.37)$$

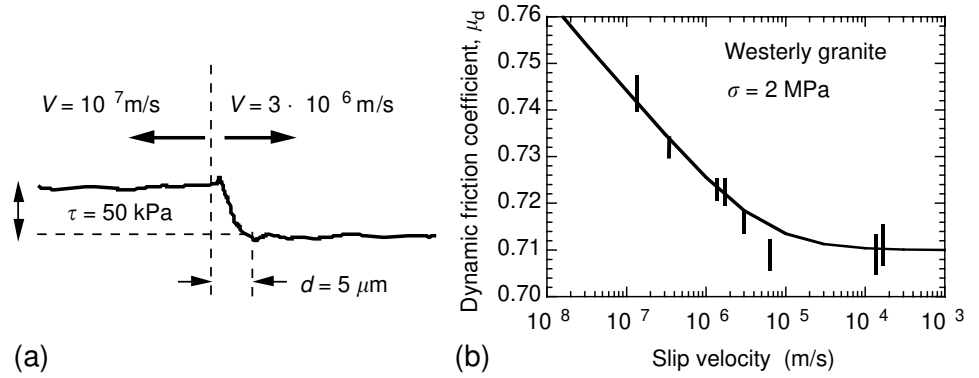
which reduces to a form essentially equivalent to (3.36) for large  $t$ , but gives  $\mu \approx \mu_0$  for  $t \ll t_0$ . Dieterich found that, for all of his data sets,  $t_0 \approx 1$  s. Subsequent investigations by Scholz et al. (1972), Engelder et al. (1975) and others have shown that an increase in the coefficient of static friction with increasing stick time is a general characteristic of rock behavior under a variety of test conditions and surface properties.

The increase in the friction coefficient with time has been attributed to “indentation creep,” which causes an increase in the actual area of asperity contact between the two contacting surfaces. Scholz and Engelder (1976) carried out microindentation tests on a natural quartz crystal and an olivine crystal, by

**Fig. 3.10** (a) Dependence of the coefficient of static friction of quartz sandstone on the time of stick, for two different values of the normal stress (Dieterich, 1978). (b) Variation of the actual area of contact under a pyramidal indenter, as a function of loading time (Scholz and Engelder, 1976). Vertical lines indicate the range of values observed over twenty-five measurements.



**Fig. 3.11** (a) Variation of shear stress with displacement, before and after an abrupt change in slip velocity. (b) Variation of the coefficient of dynamic friction with slip velocity for Westerly granite at a normal stress of 20 MPa. Short vertical lines indicate the range of values observed at each given velocity; solid line is from (3.38), with  $\mu_o = 0.71$ ,  $t_o = 1$  s,  $d = 5 \mu\text{m}$ , and  $a = 0.0087$  (Dieterich, 1978).



pressing a pyramidal “Vickers” diamond indenter into the surfaces under a constant normal load of 0.25 N. This load corresponds to a normal stress beneath the indenter of about 10–20 MPa. Twenty-five tests were conducted on each surface, for loading times of 1, 10, 100, and 1000 s. The actual area of the indentation was found to increase linearly with the logarithm of the indentation time (Fig. 3.11b). If the same process is assumed to occur at the asperities at which rough rock surfaces are in contact, and the friction coefficient is assumed to be proportional to the actual (as opposed to the nominal, macroscopic) area of contact, then this mechanism can qualitatively explain the increase in  $\mu$  with time.

The dynamic friction coefficient acting between two rock surfaces typically exhibits a small velocity dependence (Scholz and Engelder, 1976; Dieterich, 1978), the origin of which can be explained as follows. As the two rock surfaces slide past each other, asperities that have been in contact will lose contact, and new asperities will come into contact. The mean lifetime of asperity contact is roughly given by  $t \approx d/V$ , where  $V$  is the relative velocity between the two surfaces, and  $d$  is a characteristic asperity diameter. If this asperity contact lifetime is identified with the contact time appearing in (3.37), a relationship is found between the dynamic friction coefficient and the sliding velocity:

$$\mu_d = \mu_o + a \ln[1 + (d/Vt_o)]. \quad (3.38)$$

The parameter  $d$  can also be identified with the slip distance required for  $\mu$  to stabilize to a new value after an abrupt change in the velocity, and was found by Dieterich (1978) to equal about  $5 \mu\text{m}$  for contact between two granite surfaces (Fig. 3.11a). Using this value of  $d$ , along with the values of  $\mu_o$  and  $a$  obtained from the previously discussed static friction measurements, Dieterich (1978) found that (3.38) could accurately predict the measured variation of the coefficient of dynamic friction with velocity (Fig. 3.11b).

Ruina (1983), Tullis (1988) and others have proposed adding another term to (3.38) to account for the irreversible mechanical alteration of the rock surface that occurs during sliding. The resulting “rate/state” friction law can be expressed as

$$\tau = S_o + \mu_d \sigma, \quad \mu_d = \mu_o + a \ln(V/V_o) + b\psi, \quad (3.39)$$

where  $\psi$  is a state variable that in some sense represents the damage that has occurred to the surface, and  $a$  and  $b$  are dimensionless constants that reflect the rate at which the friction coefficient varies with changes in  $\ln V$  and  $\psi$ . The state variable is assumed to change with time according to an evolution equation of the form

$$d\psi/dt = -(V/\lambda)[\psi + \ln(V/V_o)], \quad (3.40)$$

where  $\lambda$  is a characteristic slip length. The additional term in (3.39) allows the dynamic friction coefficient to vary as slip proceeds.

Although the variations in  $\mu$  during sliding are usually small, it is mainly the *relative* magnitudes of the parameters  $a$  and  $b$  that determine whether or not the slip occurs in a stable or unstable (stick-slip) manner. The analysis given in §3.4 showed that, if the dynamic friction coefficient is independent of velocity and slip distance, the slip process will be unstable if  $\mu_d < \mu$ . Now consider an interface governed by a rate/state law of the form given by (3.39) and (3.40), with slip occurring at some constant velocity  $V_1$ . The state variable  $\psi$  will have already reached its steady-state value associated with velocity  $V_1$ . This steady-state value is found by setting  $d\psi/dt = 0$  in (3.40), which gives  $\psi_{ss1} = -\ln(V_1/V_o)$ . Hence, the steady-state dynamic friction coefficient corresponding to velocity  $V_1$  is given by (3.39) as

$$\mu_{d1} = \mu_o + (a - b) \ln(V_1/V_o). \quad (3.41)$$

Now imagine that the slip velocity changes abruptly to a new value  $V_2 = V_1 + \Delta V$ . Immediately after the change in velocity,  $\psi$  will be unchanged, so (3.39) shows that  $\mu_d$  will at first increase or decrease depending on the sign of  $a$ . Experiments typically show that  $a$  is positive, which is consistent with the slight initial increase in  $\tau$  that is seen in Fig. 3.11a. Eventually, the friction coefficient will again stabilize, at a value given by

$$\begin{aligned} \mu_{d2} &= \mu_o + (a - b) \ln[(V_1 + \Delta V)/V_o] \\ &= \mu_o + (a - b) \ln(V_1/V_o) + (a - b) \ln(\Delta V/V_1) \\ &= \mu_{d1} + (a - b) \ln(\Delta V/V_1). \end{aligned} \quad (3.42)$$

By analogy with the analysis of §3.4, slip would be expected to be stable if the friction coefficient increases following an increase in the velocity. This will occur if  $b < a$ , in which case the surface is said to be “velocity-strengthening.” The more rigorous linear stability analysis given by Rice and Ruina (1983) leads to the same conclusion, for small increments in the slip velocity.

Unstable stick-slip oscillations may occur along a velocity-weakening surface, for which  $b > a$ , depending on the stiffness and mass of the system. For the model studied in §3.4, in which  $\mu_d$  was independent of velocity, the motion was found to be unstable if  $\mu_d < \mu$ , regardless of the stiffness or mass. Gu et al. (1984) analyzed the system of Fig. 3.6a, with the frictional law given by (3.39) and (3.40), but with the normal load  $N$  not necessarily taken to be  $mg$ . They found that, for small driver velocities  $V$  (using the notation of §3.4), the motion will always be unstable if

$$k < (b - a)\sigma/d, \quad (3.43)$$

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where  $\sigma$  is the normal stress acting between the mass and the surface. For stiffnesses greater than the critical value given by (3.43), the stiffness of the system will be sufficient to inhibit the onset of instability, if the driver velocity  $V$  is small. For larger driver velocities, dynamic instabilities may occur, the critical velocity increasing with the mass of the system (Gu and Wong, 1991).

### 4.1 Introduction

The classical theories of continuum mechanics have for the most part been constructed so as to be in accord with experimental observations of the behavior of metals and other man-made engineering materials. These theories therefore describe the various types of behavior observed in metals, such as linear elastic, nonlinear elastic, plastic, brittle failure, etc. Although these concepts and models can be applied to rocks, in most cases the analogous behavior of rock is much more complex, undoubtedly because of its heterogeneous and porous nature. These classical constitutive models, and the modifications needed in order to apply them to rock, are discussed in detail in Chapters 5, 7, and 9.

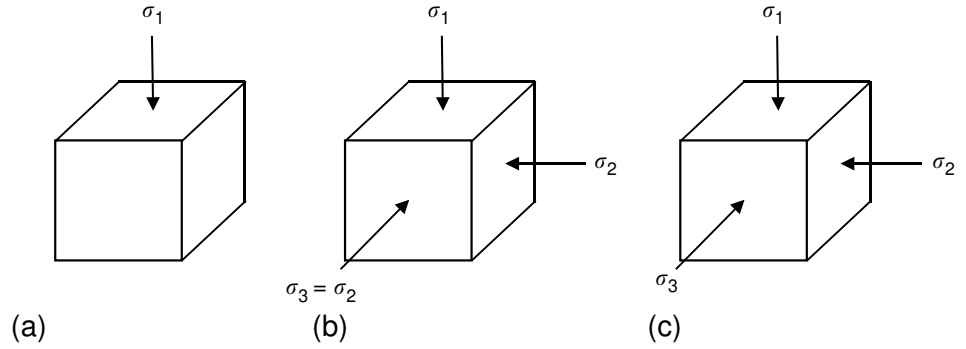
In order to develop realistic constitutive models for rock deformation, it is necessary to begin with a discussion of the types of mechanical behavior that rocks may exhibit. Hence, in this chapter we first focus on describing, qualitatively, the type of stress–strain behavior observed when rocks are subjected to external loads. The stress–strain behavior is discussed phenomenologically in §4.2 and §4.3, and from a microscopic (albeit qualitative) viewpoint in §4.4. In §4.5, the widely used Coulomb failure criterion is introduced and discussed in detail. This criterion, which assumes that failure is controlled only by the maximum and minimum principal stresses, represents a simplification of actual rock behavior, but is nevertheless extremely useful for understanding the effects of stress state on rock failure. Mohr’s generalization of Coulomb’s law, in which failure is still assumed to be governed by the two extreme principal stresses, but possibly in a nonlinear manner, is discussed in §4.6. This section also contains some discussion of one particular nonlinear failure law, that of Hoek and Brown. The effects of pore pressure, and the concept of effective stress, are discussed in §4.7. Experimental data on rocks subjected to “polyaxial,” or true-triaxial, stress states with  $\sigma_1 > \sigma_2 > \sigma_3$  are presented in §4.8 and analyzed in light of the polyaxial failure criterion proposed by Mogi. Finally, the effects of material anisotropy are briefly treated in §4.9, in terms of the “single plane of weakness” theory.

### 4.2 The stress–strain curve

The most common method of studying the mechanical properties of rocks is by axial compression of a circular cylinder whose length is two to three times



**Fig. 4.1** Cubic specimen under (a) uniaxial stress, (b) traditional triaxial stress in which the two lateral confining stresses are equal, and (c) true-triaxial stress, in which all three principal stresses are possibly different.



its diameter. If the lateral surface of the rock is traction-free, the configuration is referred to as *uniaxial compression*, or *unconfined compression* (Fig. 4.1a). In this case, the resulting state of stress in the rock is  $\{\sigma_1 > 0, \sigma_2 = \sigma_3 = 0\}$ . If tractions are applied to the lateral surfaces, the experiment is referred to as one of *confined compression*. For tests done on a circular cylinder, the stresses applied in the two orthogonal directions perpendicular to the cylinder axis are necessarily equal (Fig. 4.1b), and the resulting state of stress in the rock is  $\{\sigma_1 > \sigma_2 = \sigma_3 > 0\}$ . This state is traditionally referred to as “triaxial,” despite the fact that two of the principal stresses are equal. The more general state of stress, in which  $\{\sigma_1 > \sigma_2 > \sigma_3 > 0\}$ , can be achieved with cubical specimens, and is known either as “polyaxial” or “true triaxial” (Fig. 4.1c). The technical aspects of carrying out these experiments are discussed in Chapter 6.

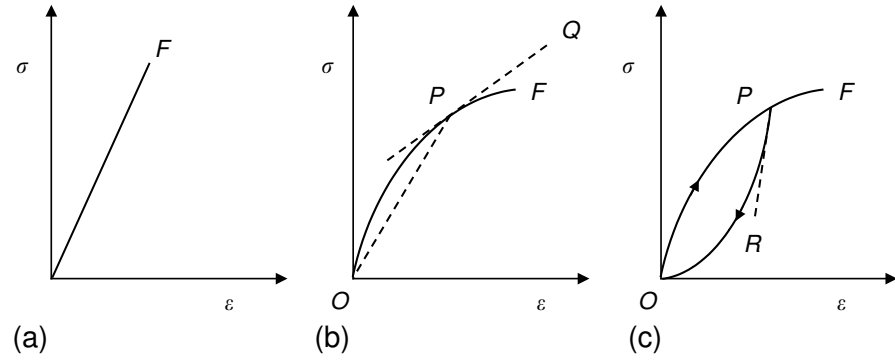
We focus our attention for now on the so-called “triaxial” test carried out on a cylindrical specimen, in which the stresses are monitored, and the axial and lateral strains are measured by means of strain gauges attached to the specimens, or by deformation gauges attached to the end-caps (see §6.4). Typically,  $\sigma_2$  and  $\sigma_3$  are held constant, while  $\sigma_1$  is increased. The results can be plotted in the form of a *stress–strain curve*, in which  $\sigma$  is plotted against  $\varepsilon$ . Strictly speaking, these variables are  $\sigma_1$  and  $\varepsilon_1$ , but the subscripts will for now be dropped for simplicity of notation.

The simplest possible behavior is illustrated in Fig. 4.2a, in which the strain increases linearly with stress, ending in abrupt failure at some point  $F$ . This curve may be represented by the equation

$$\sigma = E\varepsilon, \quad (4.1)$$

where  $E$ , which has units of Pa, is called *Young’s modulus* or the *elastic modulus*. Within the range of stress and strain prior to failure, this type of behavior is known as *linearly elastic*. The adjective “linear” refers to the mathematically linear relationship between stress and strain. In modern discussions of continuum mechanics, the term “elastic” means that the strain is a single-valued function of the stress, and does not depend on the stress history or stress path. For a linear elastic material, at any value of strain below point  $F$  in Fig. 4.2a, the slope of the curve will be given by  $d\sigma/d\varepsilon = E$ .

**Fig. 4.2** (a) Linearly elastic behavior, with failure at  $F$ . (b) Nonlinearly elastic behavior: slope of  $OP$  is the secant modulus, slope of  $PQ$  is the tangent modulus. (c) Hysteretic material: unloading modulus at  $P$  is given by the slope of  $PR$ .



By this definition of elastic behavior, a material is elastic as long as the stress can be expressed as a single-valued function of the strain, which may or may not be linear, that is,

$$\sigma = f(\varepsilon). \quad (4.2)$$

Under this definition, the stress–strain behavior of an elastic material during “loading” is the same as during “unloading.” The behavior illustrated in Fig. 4.2b is therefore referred to as *nonlinearly elastic*. For nonlinearly elastic materials, the slope of the stress–strain curve varies with the level of stress (or strain). Two types of “elastic moduli” can be defined for such materials, each of which will in general vary with both  $\sigma$  and  $\varepsilon$ . The *secant modulus* is defined to be the ratio of the total stress to the total strain, that is,

$$E_{\text{sec}} = \sigma / \varepsilon, \quad (4.3)$$

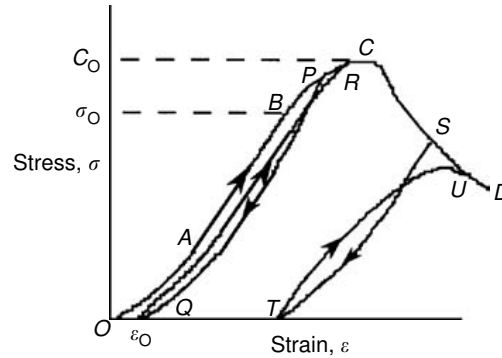
and is equal to the slope of line  $OP$ . The *tangent modulus*, on the other hand, is the local slope of the stress–strain curve,

$$E_{\text{tan}} = d\sigma / d\varepsilon, \quad (4.4)$$

and in Fig. 4.2b would be equal to the slope of line  $PQ$ . The secant and tangent moduli coincide for a linearly elastic material.

A rock is called *hysteretic* if it follows different stress–strain curves during loading and unloading, but returns to its original strain-free state when the stresses are removed. Figure 4.2c shows the stress–strain behavior of a hysteretic material, with the dashed line representing the unloading curve. Such a material exhibits a different tangent modulus during unloading than during loading; the loading modulus at  $P$  would be given by the tangent to curve  $OPF$ , whereas the unloading modulus at that same point  $P$  is given by the slope of the line  $PR$ . In §5.8, it is shown that the work done on the rock by the external loading agency during deformation, per unit volume of rock, is equal to the area under the stress–strain curve. Hence, for a rock exhibiting stress–strain behavior such as shown in Fig. 4.2c, the work done on the rock during loading would be greater than that done on it during unloading. The area between the loading and unloading portions of the stress–strain curve therefore represents energy that is

**Fig. 4.3** Complete stress–strain curve for a rock under compression (see text for details).



dissipated (by friction along grain boundaries and crack faces, etc.). As the net amount of work done *on* the rock during a cycle of loading and unloading must be nonnegative, the unloading portion of the stress–strain curve cannot lie above the loading portion.

The idealized materials described in Fig. 4.2 each deform until  $F$ , at which point they fail abruptly if the applied stress is increased further. This type of abrupt failure is observed in materials under tension, but the behavior of a rock under the more commonly occurring compressive stress regime is more complicated (Fig. 4.3). The stress–strain curve for a rock under uniaxial compression can be divided conceptually into four regions. In region  $OA$ , the curvature, roughly indicated by the second derivative, is positive. In region  $AB$ , the curve is very nearly linear. The curve continues to rise in region  $BC$ , but the curvature is now negative. The strain reaches a maximum at  $C$ , after which it falls throughout region  $CD$ .

In the first two regions,  $OA$  and  $AB$ , the behavior is nearly elastic. Some slight hysteresis may be observed, but loading and unloading in this region will not produce irreversible changes in the structure or properties of the rock. In the third region,  $BC$ , which usually begins at a stress of about two-thirds the maximum stress at  $C$ , the slope of the stress–strain curve, that is, the tangent modulus, decreases steadily to zero as the stress increases. In this region, irreversible changes occur in the rock, and successive cycles of loading and unloading would trace out different curves. An unloading cycle such as  $PQ$  that starts in region  $BC$  would lead to a permanent strain  $\varepsilon_0$  when the stress reaches zero. If the rock is reloaded, a curve such as  $QR$  would be traced out that lies below the original loading curve  $OBC$ , but which ultimately rejoins it, at a stress greater than the stress at  $P$ .

The fourth region,  $CD$ , begins at the point of maximum stress  $C$ , and has a negative slope. An unloading cycle such as  $ST$  that begins in this region would lead to a large permanent strain when the stress reaches zero. Subsequent reloading will trace out a curve in the  $\{\sigma-\varepsilon\}$  plane that rejoins the initial loading curve at  $U$ , corresponding to a stress that is *lower* than that at the beginning of the cycle, point  $S$ . In this sense we can say that the ability of the rock to support a load has decreased. This region of the stress–strain curve cannot be observed in a testing machine in which the stress is the controlled variable, in which case violent failure of the specimen will occur near point  $C$ . But this region can be

observed in a servo-controlled testing machine in which strain is the controlled variable, as discussed in detail in §6.5. This region is also of importance in a rock mass, in which the decreasing ability of one region of rock to support an additional load could be compensated for by some of the load being transferred to adjacent regions of rock.

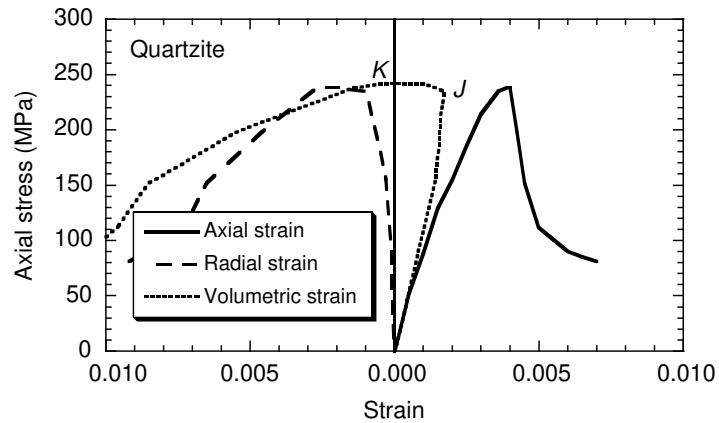
In region *BC*, the rock is said to be in a *ductile state*, or simply to be *ductile*. Ductile behavior is characterized by the ability of the rock to support an increasing load as it deforms. In region *CD*, on the other hand, the load supported by the rock decreases as the strain increases. A rock exhibiting this behavior, which is qualitatively different from the ductile behavior described above, is said to be in a *brittle state*, or simply to be *brittle*. The range of stresses in which a rock exhibits either of these two types of behavior depends on the mineralogy, microstructure, and also on factors such as the temperature, as discussed in §4.3, and in more detail by Paterson (1978).

The value of the stress at point *B*, which marks the transition from elastic to ductile behavior, is known as the *yield stress* of the rock, and is usually denoted by  $\sigma_0$ . The value of the stress at point *C*, which marks the transition from ductile to brittle behavior, is known as the *uniaxial compressive strength* of the rock, and is usually denoted by  $C_0$ . The process of failure is regarded as a continuous process that occurs throughout the brittle region *CD*, in which the rock physically deteriorates, and its ability to support a load decreases. Failure therefore begins at *C*, and the *criteria for failure* for a rock subjected to uniaxial compression would simply consist of the condition that “failure occurs when  $\sigma = C_0$ .” The failure criteria that are discussed in later sections of this chapter represent attempts to predict the onset of failure under stress states that are more general than uniaxial compression.

The foregoing discussion has shown that the axial strain that occurs under uniaxial compression can be quantified in terms of the Young’s modulus,  $E$ . But a rock under a uniaxial compressive stress will not only deform in the direction of the load, it will also deform in each of the two directions perpendicular to the load. Figure 4.4 shows the strains measured on a cylindrical specimen by Hojem et al. (1975) during confined uniaxial compression of an argillaceous quartzite, with the lateral confining stress held constant at  $\sigma_2 = \sigma_3 = 6.9$  MPa (1000 psi). As before, the convention used is that positive normal strains correspond to decreases in the linear dimensions of the specimen. The axial stress vs. axial strain curve exhibits most of the features described above, including a pronounced brittle regime. The strain in the two other directions (i.e., the radial strain,  $\varepsilon_2 = \varepsilon_3$ ) is negative, which is to say that the specimen bulges outward as it is compressed. Within the elastic regime that corresponds to region *OB* of Fig. 4.3, the magnitude of the radial strain increases nearly in proportion to the axial strain. The negative of the ratio of the transverse strain to the axial strain,  $-\varepsilon_2/\varepsilon_1$ , is known as *Poisson’s ratio*, and is denoted by  $\nu$ . For a linear elastic material, this parameter is independent of stress, and is generally found to be in the range 0–0.5 (§5.2).

In the ductile regime, corresponding to region *BC* of Fig. 4.3, the transverse strains begin to grow (in magnitude) at a much faster rate than does the axial

**Fig. 4.4** Axial strain, radial strain, and volumetric strain as a function of axial stress, for a cylindrical sample of an argillaceous quartzite, tested under a confining stress of 6.9 MPa by Hojem et al. (1975).



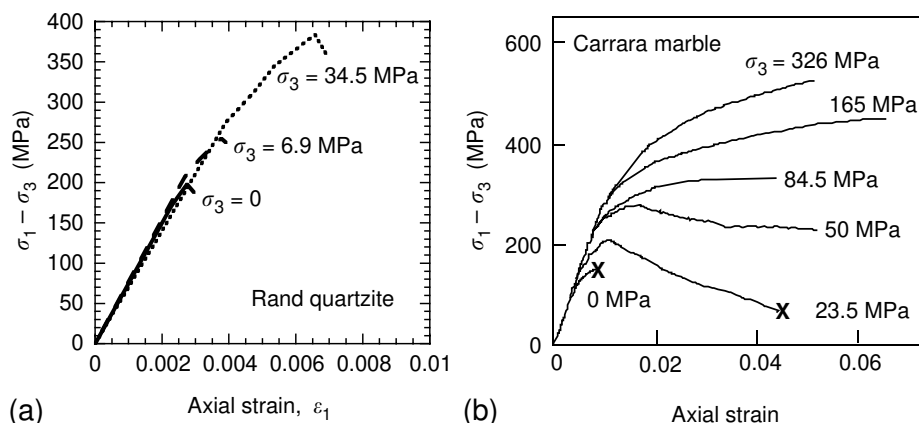
strain. In terms of incremental strains, this behavior could be said to correspond to a value of Poisson's ratio that exceeds unity. As the volumetric strain,  $\Delta V/V$ , is equal to the sum of the strains in the axial and the two lateral directions, the volumetric strain begins to decrease with an increase in the axial stress. Bearing in mind our sign convention, this means that an incremental increase in the compressive axial stress causes an incremental *increase* in the volume. This first occurs at *J* in Fig. 4.4. Eventually, the lateral strains become sufficiently negative that the *total* volumetric strain becomes negative; this occurs at *K* in Fig. 4.4. The phenomenon by which the volume of the rock decreases under the action of an additional compressive stress is known as *dilatancy*. Dilatancy can be ascribed to the formation and extension of open microcracks whose axes are oriented parallel to the direction of the maximum principal stress (§4.4). By testing specimens in the form of thick-walled hollow tubes, Cook (1970) showed that dilatancy occurs pervasively throughout the entire volume of the rock, and is not a superficial phenomenon localized at the outer boundary. On the other hand, Spetzler and Martin (1974) and Hadley (1975) have shown that dilatancy is not uniformly distributed throughout the specimen, but rather becomes increasingly heterogeneous as failure is approached.

### 4.3 Effects of confining stress and temperature

It has been known since the end of the nineteenth century that if the confining stress applied to the sides of a cylindrical specimen during a triaxial compression test is increased, the axial stress required to cause failure will increase, and the rock will show a tendency toward greater ductility (Becker, 1893; Adams, 1912). In the classical experiments performed by von Kármán (1911) and Böker (1915), oil was used to apply a confining stress  $\sigma_2 = \sigma_3$  to the sides of the specimen, while the axial stress  $\sigma_1$  was slowly increased.

The effect that confining pressure has on the axial stress *vs.* axial strain curve is shown in Fig. 4.5a for a Rand quartzite. For each value of  $\sigma_2 = \sigma_3$ , the stress-strain curve initially exhibits a nearly linear elastic portion, with a slope (Young's modulus) that is nearly independent of the confining stress. But both the yield stress and the failure stress increase as the confining stress increases. Finally, there is a small descending portion of the curve, ending in brittle fracture.

**Fig. 4.5** Stress–strain curves for (a) Rand quartzite and (b) Carrara marble at various confining pressures. Crosses indicate abrupt brittle failure.



A different type of behavior is exhibited by other rocks, notably carbonates and some sediments. Figure 4.5b shows the data collected by von Kármán (1911) on a Carrara marble. For sufficiently low confining stresses, exemplified by the curve labeled  $\sigma_3 = 0$  MPa, brittle fracture (denoted by X) occurs as for the quartzite described above. But at higher confining stresses, such as the curve labeled 50 MPa, the rock can undergo a strain as large as 7 percent, with no substantial loss in its ability to support a load (i.e., no decrease in the axial stress). In this case the rock is said to exhibit *ductile behavior*, which can be loosely defined as “the capacity for substantial change in shape without gross fracturing” (Paterson, 1978). The curve for  $\sigma_3 = 23.5$  MPa can be said to show a transitional type of behavior, in that fairly substantial inelastic strain occurs, but the rock eventually fails by brittle fracture. Hence, there is a somewhat ill-defined value of the confining stress at which one can say there occurs a transition between brittle and ductile behavior. Heard (1960) proposed that this *brittle–ductile transition* be taken as that confining stress at which the strain at failure is, say, 3–5 percent. At still higher confining stresses, such as 165 MPa or above in Fig. 4.5b, the axial stress  $\sigma_1$  continues to increase with strain after the yield point has been passed. Such behavior is known as *work hardening* in metallurgy, and more simply as *hardening* in rock mechanics. Following this nomenclature, the behavior shown in the descending portion of the stress–strain curve, such as that exhibited in Fig. 4.5b at a confining stress of 23.5 MPa, is often referred to as *softening*. Table 4.1 shows the measured brittle–ductile transition pressures for different rock types, at room temperature, as compiled by Paterson (1978) from various sources.

Higher temperatures generally have the effect, roughly speaking, of encouraging ductility. Figure 4.6a shows the stress–strain curves measured by Griggs et al. (1960) on a granite, at a confining stress of 500 MPa. At room temperature the rock is brittle, but at higher temperatures substantial amounts of permanent deformation may occur. By 800°C, the rock is almost fully ductile, in that the strain can continue to increase at a nearly constant load. Hence, for a fixed value of the constant confining stress, brittle behavior gives way to ductile behavior above a certain temperature. As both higher temperatures and higher confining stresses tend to favor ductility, the brittle–ductile transition temperature decreases as the confining stress increases. Heard (1960) developed a phase

**Table 4.1**

Brittle–ductile transition pressures under  $\sigma_2 = \sigma_3$  compression, at room temperature (after Paterson, 1978).

| Rock type              | $\sigma_2$ (b $\rightarrow$ d) (MPa) | Source(s)                                        |
|------------------------|--------------------------------------|--------------------------------------------------|
| Limestones, marbles    | 30–100                               | Heard (1960), Rutter (1972a)                     |
| Dolomite               | 100–200+                             | Handin and Hager (1957), Mogi (1971)             |
| Gypsum                 | 40                                   | Murrell and Ismail (1976)                        |
| Anhydrite              | 100                                  | Handin and Hager (1957)                          |
| Rocksalt               | <20                                  | Handin (1953)                                    |
| Talc                   | 400                                  | Edmond and Paterson (1972)                       |
| Serpentine             | 300–500                              | Raleigh and Paterson (1965)                      |
| Chloritite             | 300                                  | Murrell and Ismail (1976)                        |
| Argillaceous sandstone | 200–300                              | Edmond and Paterson (1972), Schock et al. (1973) |
| Siltstones, shales     | <100                                 | Handin and Hager (1957)                          |
| Porous lavas           | 30–100                               | Mogi (1965)                                      |

**Fig. 4.6**

(a) stress–strain curves for granite at various temperatures, at a confining pressure of 500 MPa, after Griggs et al. (1960).  
 (b) Brittle–ductile phase diagram for Solenhofen limestone, after Heard (1960). “Extension” refers to loadings for which the axial stress is less compressive than the lateral stresses.

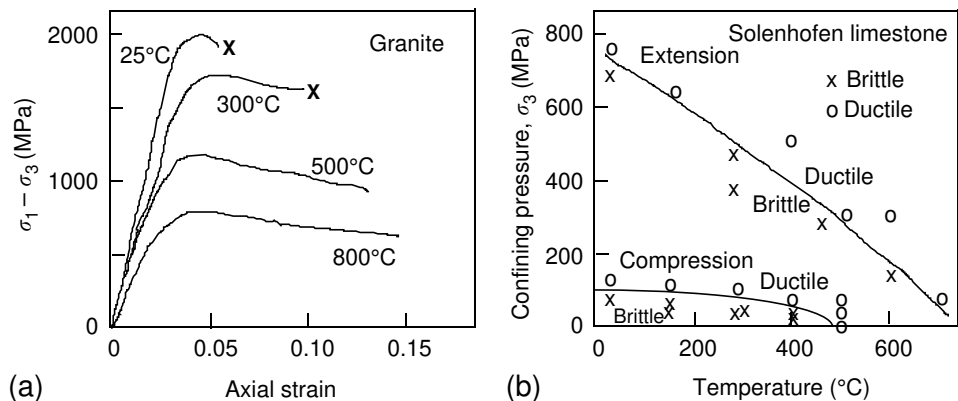


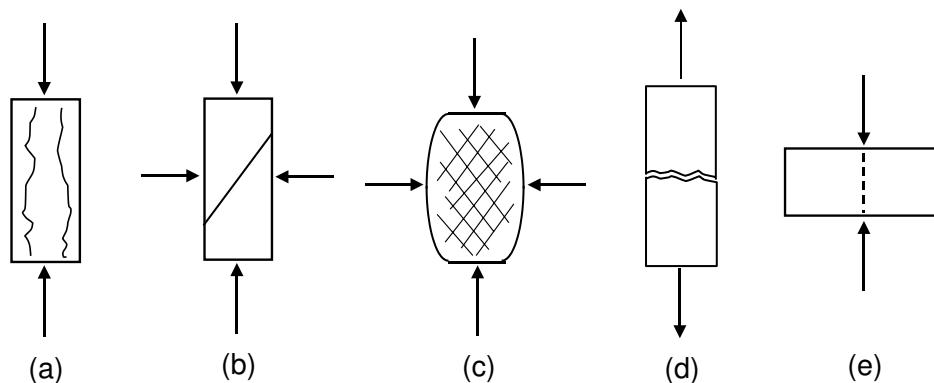
diagram for Solenhofen limestone in  $\{T, \sigma_3\}$  space, which shows regions of ductile or brittle behavior, separated by a ductile–brittle transition curve (Fig. 4.6b). For this rock, ductility can be observed under zero confining stress if the temperature is above about 500°C. For most rocks, however, if there is no confining stress the behavior will be brittle up to the melting temperature (Murrell and Chakravarty, 1973).

#### 4.4 Types of fracture

The different types of stress–strain behavior discussed in the previous two sections correspond to different physical processes occurring within the rock. Under unconfined compression, a rock tends to deform elastically, until failure occurs abruptly (Fig. 4.5). This failure is accompanied by somewhat irregular longitudinal splitting (Fig. 4.7a). With a moderate amount of confining pressure, longitudinal fracturing is suppressed, and failure occurs along a clearly defined plane of fracture (Fig. 4.7b). This plane is typically inclined at an angle less than 45° from

**Fig. 4.7**

(a) Longitudinal splitting under uniaxial tension, (b) shear fracture, (c) multiple shear fractures, (d) extension fracture, and (e) extension fracture produced by opposing line loads.



the direction of  $\sigma_1$  (the axial direction, in this case). This plane is characterized by shearing displacement along its surface, and is referred to as a *shear fracture*. Under some circumstances, failure occurs along two conjugate shear planes located symmetrically with respect to the axial direction, but this seems to be an experimental artifact caused by the ends of the specimen being constrained against rotation (Paterson, 1978, p. 18). If the confining pressure is increased, so that the rock becomes fully ductile, a network of small shear fractures appears, accompanied by plastic deformation of the individual rock grains (Fig. 4.7c).

The second basic type of fracture, an *extension fracture*, typically appears when a rock fails under uniaxial tension. The main characteristic of this type of fracture is a clean separation of the two halves of the sample, with no tangential offset between the two surfaces (Fig. 4.7d).

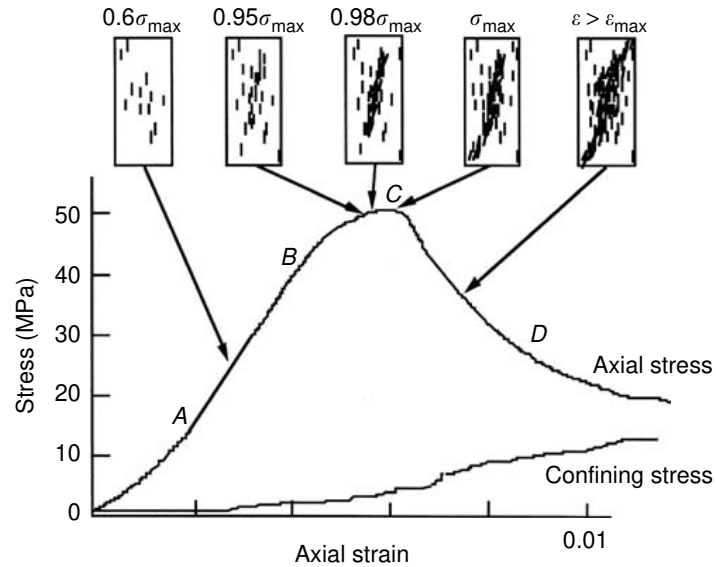
Under more complicated systems of applied stress, fractures appear which may be regarded as belonging to one or another of these two basic types. If a slab of rock is compressed between two opposing line loads (Fig. 4.7e), an extension fracture appears between the loads. If these loads are caused by a jacket surrounding the core being squeezed into cracks in the rock, the resulting fracture has been described by Brace (1964) as an *intrusion fracture*. When the fracture surfaces are examined from a specimen that has undergone longitudinal splitting, as in Fig. 4.7a, parts of the surfaces will have the appearance of a shear fracture, and other parts will appear to be extension fractures.

In §4.2, attention was directed to the phenomenon of dilatancy, which occurs during the triaxial compression of rocks. Such tests are typically conducted under conditions of constant lateral confining stress. Under such conditions, the rock is relatively free to expand laterally. In a rock mass, however, such lateral expansion would be resisted, to some extent, by the adjacent rock. One would imagine that as a portion of rock expands laterally, the lateral compressive stress imposed by the adjacent rock would increase, thereby inhibiting the lateral expansion of the rock. Hence, the deformation of a specific portion of rock *in situ* would inevitably be coupled to the deformation of the adjacent mass of rock. This coupling does not typically occur in standard rock tests, for which the boundary conditions, be they ones of constant lateral stress or constant lateral strain, are imposed a priori.

In order to approximate more closely the situation that might occur *in situ*, Hallbauer et al. (1973) jacketed specimens of a fine-grained, argillaceous



**Fig. 4.8** Schematic representation of the axial stress and lateral confining stress measured by Hallbauer et al. (1973) on a set of argillaceous quartzite specimens, along with cartoons of the state of microcracking observed on specimens that were loaded to the indicated points along the stress-strain curve.



quartzite in a copper tube of 1 mm wall thickness, and tested them in a triaxial cell in which the confining stress was applied by a pressurized fluid. An initial confining stress of 100 bars was imposed on the specimens. Calibration of the system revealed that, due to the stiffness of the cell and the small volume of confining fluid, the confining pressure increased in proportion to the lateral strain of the specimen, that is,  $\Delta\sigma_3 = c \Delta\epsilon_3$ , with the constant of proportionality found to be 1.122 GPa. Hence, the rock specimen can be thought of as being connected along its sides to linear springs of constant stiffness. The tests were conducted on a suite of specimens cut from a single block of quartzite, and were stopped at predetermined points along the stress-strain curve. Careful observation of longitudinal sections cut through the axes of the specimens allowed the growth of microcracks and fractures to be observed in relation to the stress-strain curve.

The results are illustrated in Fig. 4.8. In region AB of the stress-strain curve, the first visible structural damage appears as elongated microcracks having their axes oriented parallel (within  $\pm 10^\circ$ ) to the direction of maximum compressive stress (i.e., axially). The cracks were distributed throughout the sample, but were concentrated in the center. Toward the end of region BC, the number of microcracks increased drastically, and the cracks began to coalesce along a plane located in the central region of the specimen. At the point C of maximum axial stress, the microcracks begin to link up to form a macroscopic fracture "plane." Finally, in region CD, the fracture plane has extended through the entire specimen, and shear displacement begins to occur across the two faces of rock. In this region, the axial load carried by the specimen decreases as the rock continues to compress.

Measurements of the microcracks made after the specimens had been unloaded and sectioned showed them to be about  $300 \mu\text{m}$  long and about  $3 \mu\text{m}$  wide in their unloaded state. Their width under stress was presumably much greater than when unloaded. At each stage along the stress-strain curve, the

total volume of cracks (as measured in their unloaded state) amounted to about 16–19 percent of the inelastic volumetric dilatancy that was observed during the loading of the specimen. Hallbauer et al. (1973) concluded that the dilatant volume change reflected the opening up of these microcracks.

## 4.5 Coulomb failure criterion

The simplest, and still most widely used, failure criterion is that of Coulomb (1773). Based on his extensive experimental investigations into friction, Coulomb assumed that failure in a rock or soil takes place along a plane due to the shear stress  $\tau$  acting along that plane. In analogy with sliding along nonwelded surfaces (§3.3), motion is assumed to be resisted by a frictional-type force whose magnitude equals the normal stress  $\sigma$  acting along this plane, multiplied by some constant factor  $\mu$ . But in contrast to sliding along nonwelded surfaces, motion along the initially intact failure plane is assumed also to be resisted by an internal cohesive force of the material. Such a force reflects the fact that, in the absence of a normal stress, a finite shear stress,  $S_o$ , is typically still needed in order to initiate failure. These considerations lead to the mathematical criterion that failure will occur along a plane if the following condition is satisfied:

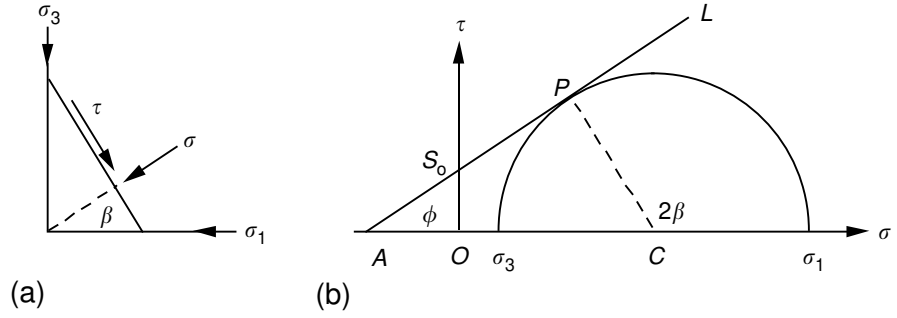
$$|\tau| = S_o + \mu\sigma. \quad (4.5)$$

(The sign of the shear stress only effects the direction of sliding after failure, so the absolute value of  $\tau$  appears in the failure criterion, although it is often convenient to ignore the absolute value signs in mathematical manipulations.) Conversely, failure will not occur on any plane for which  $|\tau| < S_o + \mu\sigma$ . The parameter  $S_o$ , also sometimes denoted by  $c$ , is known as the *cohesion*. The parameter  $\mu$  is known as the *coefficient of internal friction*, as it applies along an imaginary surface that is internal to the rock before failure occurs. Although the term *coefficient of “internal” friction* derives from the mathematical analogy between (4.5) and (3.6), Savage et al. (1996) have argued that this effect is indeed due to sliding frictional forces acting along those microscale portions of the fracture surface that are not actually intact.

The form of criterion (4.5) suggests that the Mohr’s circle construction will be useful in its analysis. Indeed, (4.5) defines a straight line on the  $\{\sigma, -\tau\}$  plane that intercepts the  $\tau$ -axis at  $-S_o$ , and has slope  $\mu$ . The angle  $\phi$  that this line makes with the  $\sigma$ -axis is given by  $\phi = \tan^{-1} \mu$ , and is known as the *angle of internal friction*. We temporarily assume a two-dimensional situation, but denote the minimum principal stress by  $\sigma_3$  rather than  $\sigma_2$ , in preparation for generalizing the discussion to three dimensions. A stress state whose Mohr’s circle lies *below* the line  $AL$  in Fig. 4.9b will not give rise to failure on any plane. If the principal stresses are such that the circle *touches* the failure line, the rock will fail in shear (Fig. 4.9b). Circles that extend *above* the failure line have no meaning in this context, since, if the stresses are assumed to increase slowly starting from some “safe” stress state that lies below the line, failure will occur as soon as the Mohr’s circle first touches the line.

The point  $P$  at which the circle is tangent to the Coulomb line represents the stress state on the plane of failure. Hence, the angle by which the failure plane

**Fig. 4.9** (a) Normal and shear tractions on a plane whose outward normal is rotated from the  $\sigma_1$  direction by an arbitrary angle  $\beta$ . (b) Mohr diagram, with failure curve (4.5) shown as line  $AL$ . Failure will occur on a specific plane whose angle  $\beta$ , demarcated by line  $CP$ , is given by (4.6).



is oriented to the  $\sigma_1$  direction is given by one-half of the angle  $2\beta$  that line  $CP$  makes with the horizontal axis. By considering the intersection of line  $CP$  with the horizontal axis, it is seen that  $2\beta = 180^\circ - \angle ACP$ . From triangle  $CPA$ , it follows that  $\angle ACP = 180^\circ - \angle CPA - \angle PAC$ . Hence,  $2\beta = \angle CPA + \angle PAC$ . Since  $P$  is the point of tangency of the circle and the failure line, line  $CP$  is perpendicular to the failure line, and so  $\angle CPA = 90^\circ$ . Finally,  $\angle PAC = \phi$ , where  $\phi$  is the angle of internal friction. Hence,

$$2\beta = 90^\circ + \phi, \quad \text{or} \quad \beta = 45^\circ + \frac{1}{2}\phi. \quad (4.6)$$

Recalling that the failure criterion (4.5) involves the absolute value  $|\tau|$ , it follows that a failure line rotated clockwise by angle  $\phi$  from the horizontal could also be drawn in Fig. 4.9b, implying that the angle  $-\beta$  must also represent a possible failure plane. As the angles on the Mohr diagram represent the normal vectors to the associated planes, we conclude that there are two possible planes of shear failure, each oriented at an acute angle of  $\beta = 45^\circ - (\phi/2)$  to the maximum principal stress. These two directions are referred to as the *conjugate directions* of shear failure.

The failure criterion (4.5) can also be written in many seemingly different but equivalent forms, each of which is convenient in certain circumstances. Figure 4.9b shows that  $|CP| = (|AO| + |OC|) \sin \phi$ , which can be written as

$$\frac{1}{2}(\sigma_1 - \sigma_3) = \left[ S_0 \cot \phi + \frac{1}{2}(\sigma_1 + \sigma_3) \sin \phi \right] = S_0 \cos \phi + \frac{1}{2}(\sigma_1 + \sigma_3) \sin \phi. \quad (4.7)$$

In terms of the “two-dimensional mean stress”  $\sigma_m$ , and the maximum shear stress  $\tau_m$ , which are given by

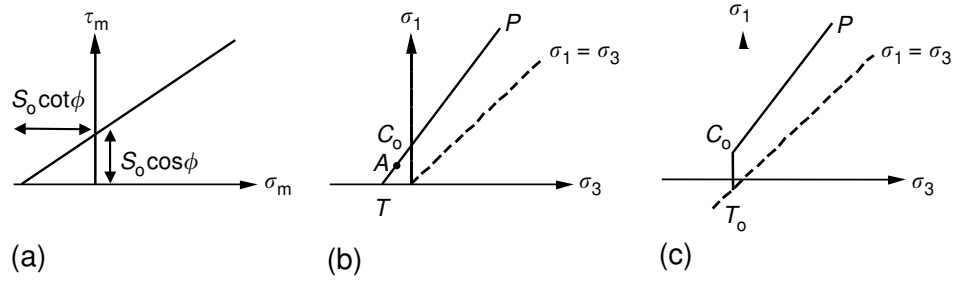
$$\sigma_m = \frac{1}{2}(\sigma_1 + \sigma_3), \quad \tau_m = \frac{1}{2}(\sigma_1 - \sigma_3), \quad (4.8)$$

the failure criterion (4.7) can be expressed as

$$\tau_m = S_0 \cos \phi + \sigma_m \sin \phi. \quad (4.9)$$

In the  $\{\sigma_m, \tau_m\}$  plane, this equation appears as a straight line that makes an angle  $\tan^{-1}(\sin \phi)$  with the  $\sigma_m$ -axis, and intercepts that axis at  $-S_0 \cot \phi$  (Fig. 4.10a).

**Fig. 4.10** (a) Failure curve in the  $\{\sigma_m, \tau_m\}$  plane, from (4.9). (b) Failure curve in the  $\{\sigma_3, \sigma_1\}$  plane, from (4.13), where region TA corresponds to *tensile* normal traction along the ostensible failure plane. (c) Same as (b), with the addition of a tension cutoff at  $T_0$ .



It would be convenient to express the Coulomb failure criterion directly in terms of the principal stresses  $\{\sigma_1, \sigma_3\}$ . First, (4.7) is rearranged as

$$\sigma_1 = 2S_0 \frac{\cos \phi}{1 - \sin \phi} + \sigma_3 \frac{1 + \sin \phi}{1 - \sin \phi}. \quad (4.10)$$

Making use of (4.6), the coefficient of  $\sigma_3$  can be transformed as follows:

$$\frac{1 + \sin \phi}{1 - \sin \phi} = \frac{1 + \sin(2\beta - 90^\circ)}{1 - \sin(2\beta - 90^\circ)} = \frac{1 - \cos 2\beta}{1 + \cos 2\beta} = \frac{2 \sin^2 \beta}{2 \cos^2 \beta} = \tan^2 \beta. \quad (4.11)$$

Similarly, the coefficient of the term  $2S_0$  can be transformed as follows:

$$\frac{\cos \phi}{1 - \sin \phi} = \frac{\cos(2\beta - 90^\circ)}{1 - \sin(2\beta - 90^\circ)} = \frac{\sin 2\beta}{1 + \cos 2\beta} = \frac{2 \sin \beta \cos \beta}{2 \cos^2 \beta} = \tan \beta. \quad (4.12)$$

Hence, the Coulomb failure criterion can also be expressed as

$$\sigma_1 = 2S_0 \tan \beta + \sigma_3 \tan^2 \beta \equiv C_0 + \sigma_3 \tan^2 \beta, \quad (4.13)$$

where  $C_0 = 2S_0 \tan \beta$  is the uniaxial compressive strength. In the  $\{\sigma_3, \sigma_1\}$  plane, the Coulomb failure criterion therefore appears as a straight line with slope  $\tan^2 \beta$ , which intercepts the  $\sigma_3$ -axis at  $-2S_0 \cot \beta$  (Fig. 4.10b).

The various forms of the Coulomb failure criterion can also be written in terms of the coefficient of internal friction,  $\mu = \tan \phi$ . Elementary trigonometry shows that

$$\cos \phi = 1/(1 + \mu^2)^{1/2}, \quad \sin \phi = \mu/(1 + \mu^2)^{1/2}, \quad (4.14)$$

which then allows (4.9) to be written as

$$(1 + \mu^2)^{1/2} \tau_m = S_0 + \mu \sigma_m. \quad (4.15)$$

Similarly, starting from (4.10) or (4.13), and making use of (4.14), the failure criterion can also be expressed in the forms

$$[(1 + \mu^2)^{1/2} - \mu] \sigma_1 = 2S_0 + [(1 + \mu^2)^{1/2} + \mu] \sigma_3, \quad (4.16)$$

$$\sigma_1 = 2S_0 [(1 + \mu^2)^{1/2} + \mu] + [(1 + \mu^2)^{1/2} + \mu]^2 \sigma_3, \quad (4.17)$$

the latter of which shows that the uniaxial compressive strength is given by

$$C_o = 2S_o[(1 + \mu^2)^{1/2} + \mu]. \quad (4.18)$$

The similarity of the foregoing discussion with that given in §3.5 should be apparent. In that section, the rock mass was assumed to have a single, preexisting plane of weakness. Alternatively, if there are no particular planes of weakness, but rather all possible planes are equally weak due to random microcracks, grain boundaries and other small-scale imperfections, the rock mass will effectively choose its own plane of failure, according to the theory outlined above. Anisotropic rock masses, on the other hand, may possess certain directions in which fracture is more likely to occur than in others; this situation is discussed in §4.9.

To simplify the preceding discussion, a two-dimensional analysis was assumed. But appealing again to the Mohr's circle construction shows that, as long as failure is assumed to occur on a plane on which the normal and shear stresses satisfy condition (4.5), consideration of a fully three-dimensional stress state alters none of the conclusions reached above. Indeed, the first Mohr's circle to touch the failure line will necessarily be that corresponding to the largest and smallest principal stresses,  $\sigma_1$  and  $\sigma_3$ . It therefore follows from Coulomb's assumption that the magnitude of the intermediate principal stress,  $\sigma_2$ , has no effect on failure. For many years, experiments could only readily be performed under the conditions  $\sigma_2 = \sigma_3$ , in which case any possible effect of the intermediate principal stress would not appear. More recent experiments under conditions of "true" triaxial conditions,  $\sigma_1 > \sigma_2 > \sigma_3$ , have revealed that this assumption is not correct. Criteria for failure under true-triaxial stress conditions are discussed in §4.8.

Coulomb's failure criterion (4.5) is essentially empirical, and thus perhaps it is not meaningful to discuss the conditions under which it is valid. Nevertheless, if it is considered that an implicit assumption contained within (4.5) is that the normal stress  $\sigma$  acting on the failure plane is positive (i.e., compressive), it would follow that certain portions of the failure curves shown in Fig. 4.10b should be ignored. The smallest value of  $\sigma_1$  that satisfies the Coulomb failure condition and that corresponds to a nonnegative normal stress on the failure plane, is found by combining the expression for  $\sigma$  with relation (4.6), to yield

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_3) + \frac{1}{2}(\sigma_1 - \sigma_3) \cos 2\beta = \frac{1}{2}(\sigma_1 + \sigma_3) - \frac{1}{2}(\sigma_1 - \sigma_3) \sin \phi > 0,$$

that is,

$$\sigma_1(1 + \sin \phi) > -\sigma_3(1 - \sin \phi). \quad (4.19)$$

Combining (4.19) and (4.10), and making use of (4.12), leads to the condition

$$\sigma_1 > S_o \tan \beta = \frac{1}{2}C_o. \quad (4.20)$$

It follows that the portion *TA* of failure line *TAP* in Fig. 4.10b does not correspond to a positive normal stress on the putative failure plane. Paul (1961)

concluded that this portion of the curve could therefore not properly predict shear failure. Indeed, this critical value of  $\sigma_1$  is positive, but (4.13) shows that it necessarily corresponds to a negative (tensile) value of  $\sigma_3$ . Experimentally, it is observed that for some negative values of  $\sigma_3$ , failure occurs by extensional fractures in planes perpendicular to this tensile stress. This mode of failure is entirely different from the shear failure that occurs under compression. The simplest modification of Coulomb's criterion that could account for this changeover in failure mode would be to truncate line  $TAP$  at point  $A$ , corresponding to  $\sigma_1 = C_o/2$  and extend it as a vertical line until it meets the line  $\sigma_1 = \sigma_3$ . (As  $\sigma_1 \geq \sigma_3$  by definition, only the region above and to the left of the line  $\sigma_1 = \sigma_3$  is meaningful.) However, this would necessarily imply that the uniaxial tensile strength,  $T_o$ , is equal to one-half the uniaxial compressive strength,  $C_o$ . Experimental data typically shows that the ratio  $C_o/T_o$  is much greater than 2. Consequently, Paul (1961) proposed using an experimentally measured value of a  $T_o$  for the location of the vertical line in the  $\{\sigma_3, \sigma_1\}$  plane (Fig. 4.10c). This is equivalent to replacing (4.13) with the bilinear failure criterion

$$\sigma_1 = C_o + \sigma_3 \tan^2 \beta, \quad \text{for } \sigma_1 > C_o[1 - C_o T_o / 4S_o^2], \quad (4.21)$$

$$\sigma_3 = -T_o, \quad \text{for } \sigma_1 \leq C_o[1 - C_o T_o / 4S_o^2]. \quad (4.22)$$

## 4.6 Mohr's hypothesis

According to Coulomb's theory, failure will occur on a plane when the normal and shear stresses acting on that plane satisfy condition (4.5). In the  $\{\sigma, |\tau|\}$  plane, this condition appears as a straight line with slope  $\mu = \tan \phi$ . The Mohr's circle corresponding to any state of stress that leads to failure will be tangent to this line. As discussed in §4.5, this theory ignores the effect of the intermediate principal stress. However, in principle, Coulomb's theory could be expected to apply to stress states in which  $\sigma_2 = \sigma_3$ . Leaving aside for now the issue of triaxiality, it is nevertheless the case that Coulomb's law is unrealistic in at least two respects.

Within the context of Coulomb's theory, the uniaxial tensile strength,  $T_o$ , can be found by setting  $\sigma_1 = 0$  in (4.17), yielding

$$T_o = -\sigma_3 = 2S_o \cot \beta = \frac{2S_o}{(\mu^2 + 1)^{1/2} + \mu}. \quad (4.23)$$

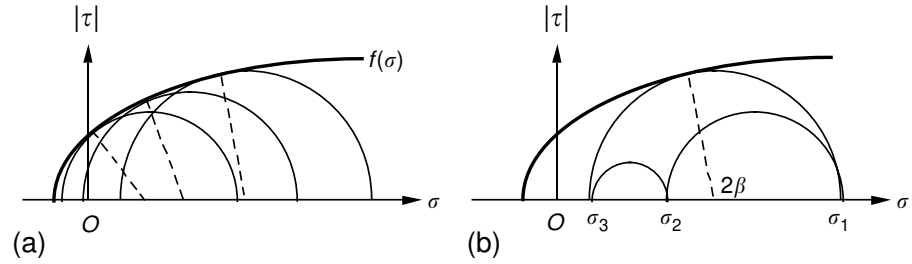
According to (4.13), the unconfined compressive strength is given by  $C_o = 2S_o \tan \beta$ . Hence, the Coulomb theory predicts that the ratio of unconfined compressive strength to unconfined tensile strength will be

$$C_o/T_o = \tan^2 \beta = [(\mu^2 + 1)^{1/2} + \mu]^2. \quad (4.24)$$

The Coulomb theory thereby predicts a relatively modest ratio of compressive to tensile strength. For example, small values of  $\mu$  lead to ratios not much larger than unity, whereas a coefficient of internal friction as large as  $\mu = 1$  leads to a strength ratio of only 5.83. Experimental values of this ratio, however, tend to be on the order of 10 or so. Roughly, this deficiency can be expressed by saying that the Coulomb failure line extends too far into the tensile region of the  $\{\sigma, |\tau|\}$  plane. This empirical observation is entirely independent of the more

**Fig. 4.11**

(a) Nonlinear failure curve, defined as the envelope of all Mohr circles that cause failure.  
 (b) Construction showing that, according to the Mohr hypothesis, the intermediate principal stress does not influence the onset of failure.



fundamental theoretical argument of Paul (1961), discussed in §4.5, regarding the supposed inapplicability of the Coulomb theory to planes on which the normal stress is tensile.

Coulomb's theory also predicts that the compressive stress required to cause failure,  $\sigma_1$ , will increase linearly with the confining stress,  $\sigma_3$ . Experiments typically show that  $\sigma_1$  at failure increases at a less-than-linear rate with  $\sigma_3$ .

In order to correct these deficiencies in Coulomb's theory, Mohr (1900) suggested that Coulomb's equation, (4.5), be replaced by a more general, possibly nonlinear, relation of the form (Fig. 4.11a)

$$|\tau| = f(\sigma). \quad (4.25)$$

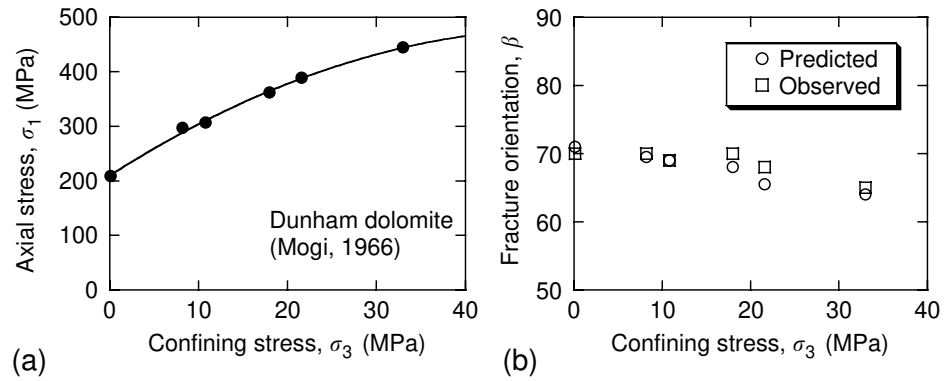
In principle, this curve can be determined experimentally as the envelope of all of the Mohr's circles that correspond to states of stress that cause failure. Aside from the fact that  $f$  may now be a nonlinear function, the basic ideas of Coulomb's model are retained. Specifically, failure is supposed to occur if one of the Mohr's circles touches the curve defined by (4.25). As shown in Fig. 4.11b, this will necessarily occur for the circle defined by  $\sigma_1$  and  $\sigma_3$ , and so the value of the intermediate principal stress is not expected to affect the onset of failure.

Moreover, as the state of stress at the point of contact of the Mohr's circle and the failure curve represents the stresses acting on the failure plane, the generalized Mohr theory predicts, as did the Coulomb theory, that the failure plane passes through the direction of the intermediate principal stress, and its normal vector makes an angle  $\beta$  with the direction of maximum principal stress, where  $2\beta$  is the angle by which line  $PC$  is rotated (counterclockwise) from the  $\sigma_1$ -axis (Fig. 4.11b). If the failure criterion (4.25) is concave downward, as is usually the case, the angle  $\beta$  of the failure plane will decrease with increasing confining stress, as indicated in Fig. 4.11a.

Experimentally, the failure criterion (4.25) can be determined by plotting the Mohr's circles for the stresses at failure, as found in a series of tests conducted under different confining stresses. The failure curve will then be given by the envelope of these circles (Fig. 4.11a). Alternatively, the stresses at failure could be plotted in the  $\{\sigma_3, \sigma_1\}$  plane, thereby generating a nonlinear analogue to (4.13). Numerous mathematical formulae, each containing two or more adjustable parameters, have been proposed for the purposes of fitting such failure data. Many of these formulae are discussed by Andreev (1995) and Sheorey (1997).

Regardless of the specific form taken by the failure curve, an unambiguous prediction of the Mohr theory is that the orientation of the plane of failure

**Fig. 4.12** (a) Stresses at failure, measured by Mogi (1966) on a Dunham dolomite. (b) Comparison of the failure angles predicted by the Mohr construction, and the observed angles (between the normal to the failure plane and the direction of maximum principal stress).



can be predicted by the construction shown in Fig. 4.11b. Data obtained under traditional  $\sigma_2 = \sigma_3$  triaxial tests typically show reasonably good agreement with this aspect of the theory. For example, Mogi (1966) conducted a series of triaxial failure tests on a Dunham dolomite, determined the orientation angles  $\beta$  from the Mohr construction, and then compared these angles to the observed angles of the failure plane. As seen in Fig. 4.12, the failure curve in the  $\{\sigma_3, \sigma_1\}$  plane was slightly nonlinear, but the observed failure angles were generally quite close to the predicted values.

All empirical failure criteria that follow Mohr's hypothesis will be expressible in some functional form  $|\tau| = f(\sigma)$ , or, alternatively,  $\sigma_1 = g(\sigma_3)$ . Either of these two functions,  $f$  or  $g$ , will suffice to determine the other. However, as will be clear from the analysis of the Coulomb theory given in §4.5, the relationship between these two functions will in general not be simple or obvious.

One such failure law that has become widely used, and which is capable of fitting data from many different rocks, is the Hoek–Brown criterion (Hoek and Brown, 1980). In terms of the two extreme principal stresses, this criterion takes the form (Fig. 4.13a)

$$\sigma_1 = \sigma_3 + (m\sigma_c\sigma_3 + \sigma_c^2)^{1/2}, \quad (4.26)$$

where  $m$  and  $\sigma_c$  are two fitting parameters. Setting  $\sigma_3 = 0$  in (4.26) shows that  $\sigma_c$  is in fact equal to the uniaxial compressive strength,  $C_o$ . Setting  $\sigma_1 = 0$  in (4.26) and solving the resultant quadratic equation for the uniaxial tensile strength  $T_o = -\sigma_3$ , gives

$$T_o = \frac{\sigma_c}{2}[(m^2 + 4)^{1/2} - m]. \quad (4.27)$$

The parameter  $m$  usually lies in the range  $5 < m < 30$ , in which case an expansion of (4.27) for “large”  $m$  leads to

$$T_o \approx \sigma_c/m, \quad \text{that is, } C_o/T_o \approx m. \quad (4.28)$$

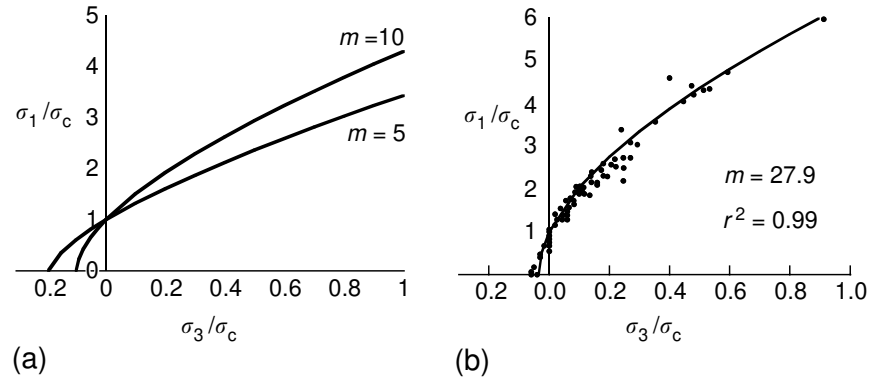
In practice, therefore, the Hoek–Brown model predicts a much larger ratio of compressive strength to tensile strength than does the Coulomb model, and so in that regard is in closer agreement with experimental data.



**Fig. 4.13**

(a) Hoek–Brown failure curves, for two different values of  $m$ .

(b) Experimental failure data for several different granites, fit with different values of  $\sigma_c$ , but one value of  $m$  (after Hoek and Brown, 1980).



Analysis of published strength data led Hoek and Brown (1980) to conclude that  $m$  takes on characteristic values for different rock types. They summarized these trends as follows:

- 1  $m \approx 7$  for carbonate rocks with well-developed crystal cleavage (dolomite, limestone, marble);
- 2  $m \approx 10$  for lithified argillaceous rocks (mudstone, siltstone, shale, slate);
- 3  $m \approx 15$  for arenaceous rocks with strong crystals and poorly developed crystal cleavage (sandstone, quartzite);
- 4  $m \approx 17$  for fine-grained polyminerallic igneous crystalline rocks (andesite, dolerite, diabase, rhyolite);
- 5  $m \approx 25$  for coarse-grained polyminerallic igneous and metamorphic rocks (amphibolite, gabbro, gneiss, granite, norite, quartz-diorite).

Figure 4.13b shows data on granites, collected from various sources by Hoek and Brown (1980) and plotted in normalized form. A single best-fit value of  $m = 27.86$  was found for the combined data set, although specific values of  $\sigma_c$  were found for each different granite in the set. The resulting fit has a correlation coefficient of 0.99.

Although the Hoek–Brown model gives a nonlinear failure envelope on a Mohr diagram, in contrast to the linear relationship predicted by the Coulomb model, Hoek (1990) has presented equations that allow the Hoek–Brown failure envelope to be locally approximated by a Coulomb line. Various technical issues associated with accurately fitting failure data to different types of failure curves are discussed by Handy (1981), Sheorey (1997), and Pincus (2000).

## 4.7 Effects of pore fluids

The foregoing discussion of rock failure has ignored the fact that rocks are typically porous to some extent, and the pore space of a rock will *in situ* be filled with fluids under pressure. The pore fluid is usually water, but may be oil, gas, or rock melt. The pore fluid may affect the failure of the rock in two ways: due to the purely mechanical effect of pore pressure, or due to chemical interactions between the rock and the fluid.

With regards to the mechanical effect of pore fluid pressure, it seems plausible that pore pressure, which acts “outward” from the pore space, would in some sense act like a tensile stress. Moreover, in an isotropic rock, this effect should be the same in any three mutually orthogonal directions. Reasoning along these lines, the soil mechanician Karl Terzaghi (1936) proposed that the failure of a soil would be controlled by the “effective stresses,”  $\sigma'_i$ , which would be the principal stresses, reckoned positive if compressive, *minus* the pore pressure, that is,

$$\sigma'_1 = \sigma_1 - P, \quad \sigma'_2 = \sigma_2 - P, \quad \sigma'_3 = \sigma_3 - P, \quad (4.29)$$

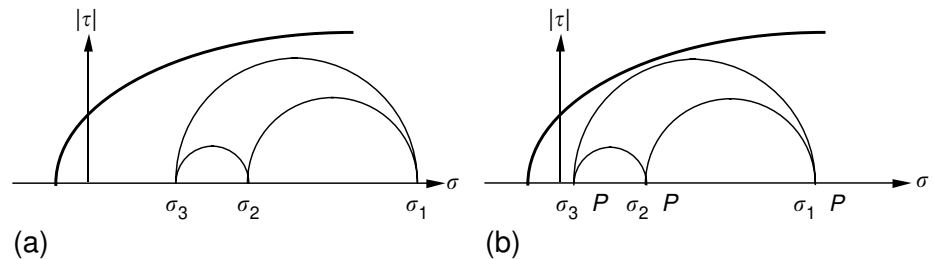
where  $P$  is the pore fluid pressure. In a more general formulation,  $P$  could be multiplied in (4.29) by some parameter  $\alpha$ , which would be referred to as the *effective stress coefficient*. (It should be emphasized that the effective stress coefficient for failure processes has no particular connection to the effective stress coefficient that appears in the theory of linear poroelasticity, §7.4.)

Most experiments on rocks support the conclusion that the effective stress law (4.29) holds, which is to say that the effective stress coefficient for failure is unity. Despite many attempts to derive an effective stress law for failure, which have been contentious and inconclusive, this “law” is best viewed as an empirical observation. It is, however, consistent with the assumption that brittle failure is in some way controlled or initiated by the stress concentrations at the corners of thin microcracks (see §10.8). As shown in §8.10, these stress concentrations are indeed proportional to the difference between the far-field stress and the fluid pressure in the crack.

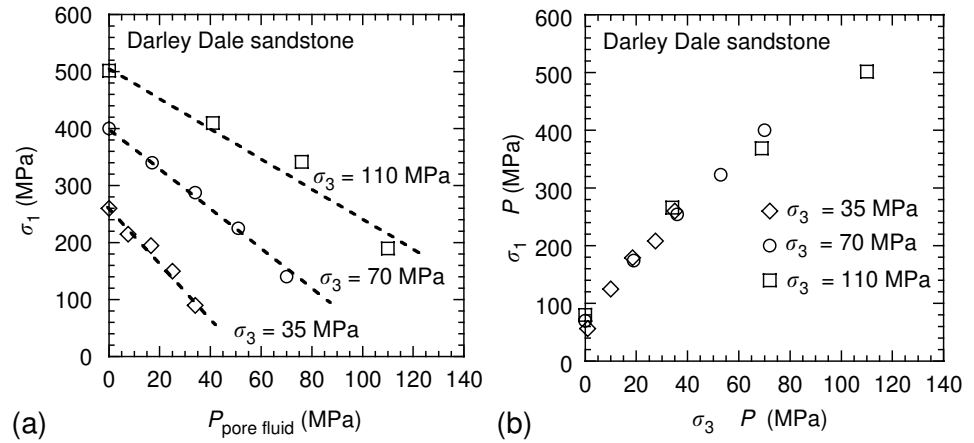
In the context of a Mohr diagram, replacing the stresses  $\sigma_i$  with the effective stresses  $\sigma'_i$  has the effect of *translating* all the stress circles to the *left* by the amount  $P$ . The Mohr’s circle will therefore be shifted closer to the failure line. Hence, an *in situ* state of stress that is “safe” in the absence of a pore pressure may well cause the rock to fail if the pore pressure is increased by a sufficient amount (Fig. 4.14). This fact accounts for the increased occurrence of landslides in the aftermath of heavy rainfall.

A set of data that illustrates the effective stress principle of brittle failure is that of Murrell (1965), who conducted standard triaxial compression tests on a Darley Dale sandstone, at several different values of the pore pressure. The Darley Dale is a poorly graded feldspathic sandstone with 21 percent porosity. In each test, the pore pressure and the confining stress were held constant, while the axial stress was increased until failure occurred. When plotted in the  $\{P, \sigma_1\}$  plane, the data fall on different curves, corresponding to the different values of  $\sigma_3$  (Fig. 4.15a).

**Fig. 4.14** (a) A stress state that lies below the failure curve. (b) Application of a pore pressure  $P$  causes the *effective* stress state to move closer to the failure curve.



**Fig. 4.15** (a) Stresses at failure in a Darley Dale sandstone (Murrell, 1965) as a function of pore pressure, for several different values of the confining stress. (b) Same data plotted in terms of the effective principal stresses, according to the effective stress law (4.29).



If plotted on the  $\{\sigma_3, \sigma_1\}$  plane, the data would form three distinct vertical lines. However, when plotted in the  $\{P, \sigma_1\}$  plane of maximum and minimum effective stresses, the failure data nearly form a single failure curve, which in this particular case is slightly concave downward (Fig. 4.15b).

It follows from (4.29) that the effective stress tensor  $\sigma'$  is related to the actual stress tensor  $\sigma$  by

$$\sigma' = \sigma - P\mathbf{I}, \quad (4.30)$$

where  $\mathbf{I}$  is the identity tensor. The additional hydrostatic stress  $-P\mathbf{I}$  gives rise to no shear stresses on any plane (§2.8), so the effective shear stresses are identical to the actual shear stresses. One implication of this fact is that the condition for sliding along a fault or other plane of weakness, (3.20), is replaced, in the presence of a pore fluid, by

$$|\tau| = S_o + \mu(\sigma - P). \quad (4.31)$$

A simple physical interpretation of this condition is that, while the normal stress  $\sigma$  tends to strengthen the fault, by pushing the two opposing rock faces together, the fluid pressure acts to weaken it, by pushing the two opposing rock faces apart. Expression (4.31) has been verified experimentally by Byerlee (1967) on laboratory specimens of granite. Some geological implications of this concept have been discussed by Secor (1965) and Hubbert and Rubey (1959, 1960, 1961).

Extension failure in the presence of pore pressure has been discussed theoretically by Murrell (1964). The effective stress concept suggests that the criterion for failure should be

$$\sigma'_3 = \sigma_3 - P = -T_o. \quad (4.32)$$

This result is consistent with the concept that tensile failure is caused by the stress concentrations at the edges of thin cracks oriented normal to the direction of the least compressive principal stress. Jaeger (1963) performed tensile failure tests on a fine-grained Tasmanian dolerite, and found that (4.32) held, although the data

was fit best by multiplying the pore fluid pressure  $P$  by an effective stress coefficient of 0.95. The slight discrepancy between the experimental and theoretical effective stress coefficients was attributed to experimental error. Condition (4.32) for tensile failure is fundamental to the theory of hydraulic fracturing (§13.6).

The strength of rocks can also be influenced by *chemical* interactions between the rock and the pore fluid (Paterson, 1978, pp. 78–9). In many quartz-rich rocks, but also in limestones, it has been found that the strength decreases if the rock is in contact with water. This effect has been noted for sandstones by Jaeger (1943), for coal by Price (1960), for calcite rocks by Rutter (1972b), and for limestone by Parate (1973). Vukoturi (1974) measured the tensile strength of Indiana limestone with different pore fluids, including water, glycerine, benzene, and various alcohols and found that the strength decreased with an increase in surface tension. The tensile strength of the limestone when saturated with water was about 25 percent less than when saturated with ethyl alcohol and about 30 percent less than the value that could be extrapolated from the data for a hypothetical pore fluid with zero surface tension.

Pore fluids can also influence rock strength through the mechanism of *stress corrosion fracture* (Atkinson, 1979; Peck, 1983). In quartz-rich rocks, fracture proceeds through the rupture of Si–O bonds at the crack tip, and this rupture is accelerated when the bonds are strained by external stresses (Dove, 1995). The rate of crack growth can be modeled as a rapidly increasing function of the applied far-field stress, such as an exponential or power law function (Wiederhorn and Boltz, 1970; Lawn, 1993). Ojala et al. (2003) conducted standard triaxial compression tests on a Lochabriggs sandstone composed of 88 percent quartz and 6 percent K-feldspar, while flowing water through the sample at a rate of 30 ml/h. The concentration of silica in the effluent was found to correlate with the different stages of the deformation process: crack closure, linear elastic, and strain hardening. The yield stress increased with increasing strain rate, consistent with the idea that at higher strain rates the fluid has less time to react with the rock. The effects of rock/fluid interactions on rock failure have been reviewed by Atkinson and Meredith (1987).

## 4.8 Failure under true-triaxial conditions

Mohr's theory of failure is based on the assumption that failure is controlled by the minimum and maximum principal stresses, and is unaffected by the magnitude of the intermediate principal stress. As most "triaxial" rock testing is conducted under conditions of  $\sigma_2 = \sigma_3$ , such data do not allow a test of this aspect of Mohr's hypothesis. Nevertheless, there are situations in which  $\sigma_2 = \sigma_3$  will be the exception in the subsurface rather than the rule. Hence, the question of whether or not Mohr's hypothesis is correct is pertinent.

Compression tests conducted under true-triaxial conditions (Mogi, 1971; Fig. 4.16a) and borehole breakout tests (Haimson and Song, 1995; Fig. 4.17a) in fact show that for many, although not all, rocks the intermediate principal stress has a pronounced influence on the value of  $\sigma_1$  at failure. This suggests the need for failure criteria that depend on all three principal stresses. As it is known from traditional ( $\sigma_2 = \sigma_3$ ) compression tests that a lateral confining stress of

the form  $\sigma_2 = \sigma_3$  has the effect of strengthening the rock, it is plausible that any increase of  $\sigma_2$  above  $\sigma_3$  may cause additional strengthening. This suggests replacing the failure criterion of the form  $\sigma_1 = f(\sigma_2)$  by the more general form

$$\sigma_1 = f(\sigma_2, \sigma_3). \quad (4.33)$$

For an isotropic rock, such a failure criterion can also always be written in terms of the stress invariants (§2.8), and it is often convenient to do so.

In the context of metal plasticity, Nadai (1950) suggested that the “driving force” for failure will be  $J_2$ , the second invariant of the deviatoric stress. According to (2.164), this invariant is related to the three principal stresses by

$$J_2 = \frac{1}{6}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]. \quad (4.34)$$

According to (5.152), in a linearly elastic, isotropic material,  $J_2$  is also a direct measure of the distortional strain energy. Nadai further suggested that failure was “opposed” by the mean stress in the material, which is consistent with the concept that confinement strengthens the rock. The mean stress can be represented by the first invariant of the stress,  $I_1$ , which according to (2.143) and (2.153) is related to the three principal stresses by

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3 = 3\tau_m, \quad (4.35)$$

where  $\tau_m$  is the mean normal stress. Hence, Nadai’s assumption can be written in the form

$$J_2 = f(I_1), \quad (4.36)$$

where  $f$  is some *increasing* function of  $I_1$ . A factor of  $2/3$  is often included in front of  $J_2$ , in which case, according to (2.167), the failure criterion can be written in terms of the octahedral shear stress and the mean normal stress:

$$\tau_{\text{oct}}^2 \equiv \frac{2}{3}J_2 = f(\tau_m), \quad (4.37)$$

where  $f$  is some increasing function of the mean normal stress.

Many specific forms of a true-triaxial failure criterion have been proposed for rocks and soils. Drucker and Prager (1952) took the relationship between  $J_2$  and  $I_1$  at failure to be of the form

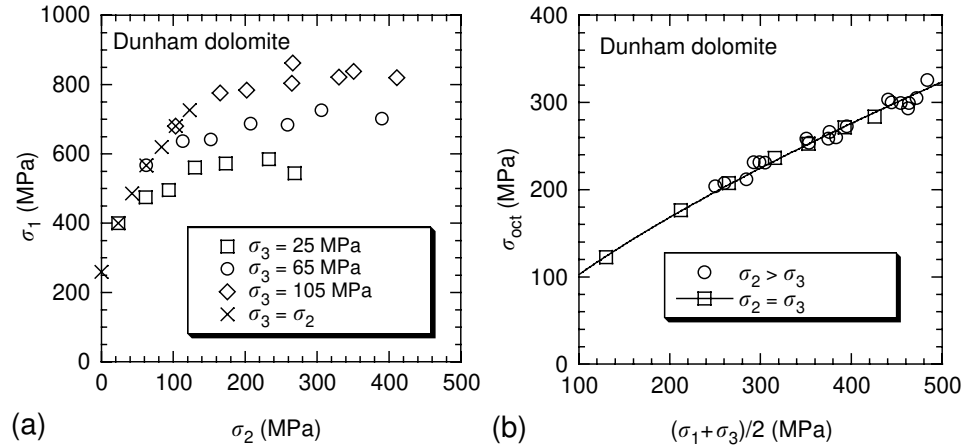
$$(J_2)^{1/2} = a + bI_1, \quad (4.38)$$

where  $a$  and  $b$  are material-dependent constants. Zhou (1994) extended this by adding a term that is quadratic in  $I_1$ :

$$(J_2)^{1/2} = a + bI_1 + cI_1^2. \quad (4.39)$$

Colmenares and Zoback (2002) referred to this as the “modified Wiebols and Cook” model in recognition of its similarities to the criterion proposed by Wiebols and Cook (1968) based on micromechanical analysis of sliding cracks (§10.6).

**Fig. 4.16** (a) Stresses at failure in a Dunham dolomite (Mogi, 1971). If failure did not depend on  $\sigma_2$ , the values of  $\sigma_1$  for fixed  $\sigma_3$  would lie on horizontal lines. (b) Same data plotted in the  $\{\tau_{m2}, \tau_{oct}\}$  plane, fit with a power law function.



Mogi (1971) modified the reasoning leading to (4.36) or (4.37) by the following argument. According to the Mohr hypothesis, the failure plane will lie parallel to the direction of the intermediate principal stress. Hence, it is plausible that fracture is resisted only by the mean normal stress in the plane normal to the failure plane, which is  $(\sigma_1 + \sigma_3)/2$ , rather than by the total mean normal stress. This suggests a brittle failure criterion of the general form

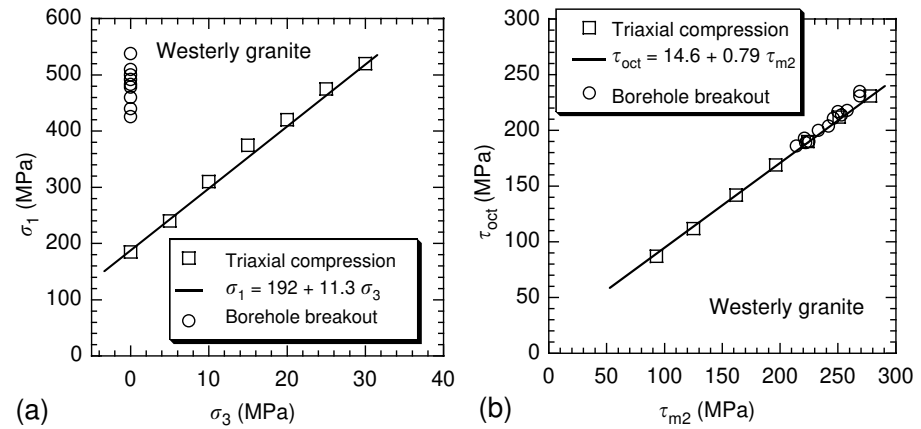
$$\tau_{oct} = f(\tau_{m2}), \quad \text{where } \tau_{m2} = (\sigma_1 + \sigma_3)/2. \quad (4.40)$$

To test this hypothesis, Mogi (1971) conducted true-triaxial compression tests on several rocks. The results showed that, for a fixed value of the minimum stress  $\sigma_3$ , the value of  $\sigma_1$  at failure at first increases with an increase in  $\sigma_2$ , but eventually decreases slightly as  $\sigma_2$  increases yet further (Fig. 4.16a). If the octahedral shear stress at failure is plotted against the mean stress on the plane parallel to  $\sigma_2$ , as suggested by (4.40), the results do indeed coalesce to a single line in the  $\{\tau_{m2}, \tau_{oct}\}$  plane (Fig. 4.16b).

Haimson and Song (1995) compared the results of standard  $\sigma_2 = \sigma_3$  confined compression tests with borehole breakout data measured on 10-cm cubical specimens containing central circular boreholes with a radius of 1 cm. The cubical specimens were subjected to various true-triaxial stress states, with the intermediate “far-field” principal stress always aligned parallel to the borehole. The local stresses at the borehole wall were calculated from the Kirsch solution, (8.113)–(8.115), for a circular hole in an infinite rock mass. The borehole wall was traction-free, so the minimum principal stress at the location of the borehole breakout, which in this case was the radial stress, was always zero, but  $\sigma_2$  was always nonzero. The values of  $\sigma_1$  at failure that were observed in the borehole breakout tests were 2–3 times greater than would be predicted by the Coulomb failure criterion that was derived from the standard compression tests (Fig. 4.17a). However, both the confined compression data and the borehole breakout data fell on a single line in the  $\{\tau_{m2}, \tau_{oct}\}$  plane (Fig. 4.17b), in accordance with Mogi’s model. For both Lac du Bonnet and Westerly granite, the failure data could be fit with curves of the form

$$\tau_{oct} = a + b\tau_{m2}. \quad (4.41)$$

**Fig. 4.17** (a) Stresses at failure in a Westerly granite, measured by Haimson and Song (1995) under standard “triaxial” confined compression and in borehole breakout tests, plotted in the  $\{\sigma_3, \sigma_1\}$  plane. (b) Same data plotted in the  $\{\tau_{m2}, \tau_{oct}\}$  plane.



The best-fitting values were found to be  $\{a = 19.0 \text{ MPa}, b = 0.76\}$  for Lac du Bonnet, and  $\{a = 14.6 \text{ MPa}, b = 0.79\}$  for Westerly.

Al-Ajmi and Zimmerman (2005) showed that a linear expression for the failure criterion in Mogi's  $\{\tau_{m2}, \tau_{oct}\}$  space, such as is given by (4.41), reduces precisely to the Mohr–Coulomb criterion if any two of the principal stresses are equal. Thus, the linear Mogi law, which they referred to as the “Mogi–Coulomb” criterion, is in some sense a natural extension of the Mohr–Coulomb criterion into the polyaxial stress domain.

Colmenares and Zoback (2002) carried out a detailed analysis of several published true-triaxial data sets for brittle failure. Data were assembled on Solenhofen limestone, Dunham dolomite, Yuubari shale, Shirahama sandstone, and amphibolite. The data for each rock were fit to several triaxial failure models, including the Drucker–Prager criterion (4.38), Zhou's modified Wiebols and Cook criterion (4.39), and Mogi's criterion (4.40) in the specific form  $f(\tau_{m2}) = c\tau_{m2}^n$ , where  $c$  and  $n$  are fitting parameters. The linear Coulomb criterion (§4.5) and the non-linear Hoek–Brown criteria (§4.6), which do not attempt to account for the effect of  $\sigma_2$ , were also used. In general, the modified Wiebols and Cook model was able to fit the data reasonably well, as was Mogi's model, whereas the Drucker–Prager model gave very poor fits. The Shirahama sandstone and Yuubari shale showed very weak dependence on the intermediate principal stress, and consequently these rocks could be fit by the Coulomb and Hoek–Brown criteria. These latter models did not provide as good fits for the other three rocks, which showed stronger dependence on  $\sigma_2$ . Although measured data could always be fit to Mogi's model, in some cases this model seemed to give multivalued predictions of the value of  $\sigma_1$  at failure, at fixed values of  $\{\sigma_2, \sigma_3\}$ , which could be problematic when used as a predictive tool.

#### 4.9 The effect of anisotropy on strength

Since most sedimentary and metamorphic rocks are anisotropic, the effect of anisotropy on strength is of great importance. The simplest situation, that of planar anisotropy in which a rock mass has a set of parallel planes of weakness, can now be addressed, by combining the results of §3.5 and §4.5. In §3.5, the problem of sliding along a preexisting plane of weakness was discussed. If  $S_w$  is

the inherent shear strength of the planes of weakness, and  $\mu_w$  is the coefficient of internal friction along those planes, then the condition for sliding along these planes, (3.28), can be written in the present notation as

$$\sigma_1 = \sigma_3 + \frac{2(S_w + \mu_w \sigma_3)}{(1 - \mu_w \cot \beta) \sin 2\beta}, \quad (4.42)$$

where  $\beta$  is the angle between  $\sigma_1$  and the normal to the planes of weakness.

As shown in §3.5, the value of  $\sigma_1$  required to cause failure, as given by (4.42), tends to infinity as  $\beta \rightarrow \pi/2$  or  $\beta \rightarrow \tan^{-1} \mu_w = \phi_w$ . For angles between these two values, failure will occur at a finite value of  $\sigma_1$  that varies with  $\beta$ . The minimum such value of  $\sigma_1$  is

$$\sigma_1^{\min} = \sigma_3 + 2(S_w + \mu_w \sigma_3)[(1 + \mu_w^2)^{1/2} + \mu_w], \quad (4.43)$$

which occurs at a specific angle  $\beta_w$  that is given by

$$\tan 2\beta_w = -1/\mu_w. \quad (4.44)$$

If the plane of weakness is oriented from the direction of maximum principal stress by some angle other than  $\beta_w$ , failure can still occur, but only at a value of  $\sigma_1$ , as given by (4.42), that is greater than  $\sigma_1^{\min}$ . For values of  $\beta < \phi_w$ , failure along the plane of weakness is not possible, for any value of  $\sigma_1$ . The situation is illustrated in Fig. 3.9.

Thus far, only the possibility of failure along a plane of weakness has been considered. However, failure can occur on a plane other than the preexisting planes of weakness if the Coulomb failure criterion, (4.5), is satisfied, but with parameters that can be denoted as  $S_o$  and  $\mu_o$ . As the planes of weakness are by definition weaker than the intact rock, it can be assumed that  $S_w < S_o$  and  $\mu_w < \mu_o$ . According to (4.13), failure can occur on a plane other than a plane of weakness if  $\sigma_1$  reaches the value

$$\sigma_1 = 2S_o \tan \beta_o + \sigma_3 \tan^2 \beta_o, \quad (4.45)$$

where, according to (4.6),  $\beta_o$  is given by

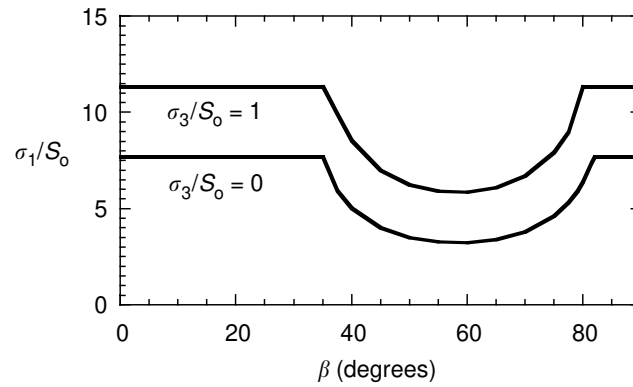
$$\tan 2\beta_o = \tan(\phi_o + 90^\circ) = -1/\tan \phi_o = -1/\mu_o. \quad (4.46)$$

For a fixed value of  $\sigma_3$ , the value of  $\sigma_1$  required to cause failure somewhere within the rock will then be equal to the *smaller* of two values given by (4.42) and (4.45). If, for a given orientation  $\beta$  of the normal to the planes of weakness relative to the direction of maximum principal stress, the value given by (4.42) is less than that given by (4.45), failure will occur along a plane of weakness. On the other hand, if the value given by (4.42) is greater than that given by (4.45), failure will occur along a plane within the intact rock whose orientation is defined by (4.46).

The value of  $\sigma_1$  needed to cause failure is plotted in Fig. 4.18, for two values of  $\sigma_3$ . The concave-upward portions of the curves correspond to criterion (4.42), and represent failure along the plane of weakness. The horizontal portions of the



**Fig. 4.18** Variation of the value of  $\sigma_1$  needed to cause failure, as a function of the angle  $\beta$  between the normal to the plane of weakness and the maximum principal stress, for the case  $\mu_w = 0.5$ ,  $\mu_o = 0.7$ ,  $S_o = 2S_w$ .



curves are plotted from (4.45), and represent failure within the intact rock, along a plane defined by (4.46). Larger values of  $\sigma_3$  would cause the curve to be shifted upward, as can be anticipated by examining Fig. 3.9 and (4.45). Experimental results (Donath, 1961; Hoek, 1964) show general agreement with that shown in Fig. 4.18.